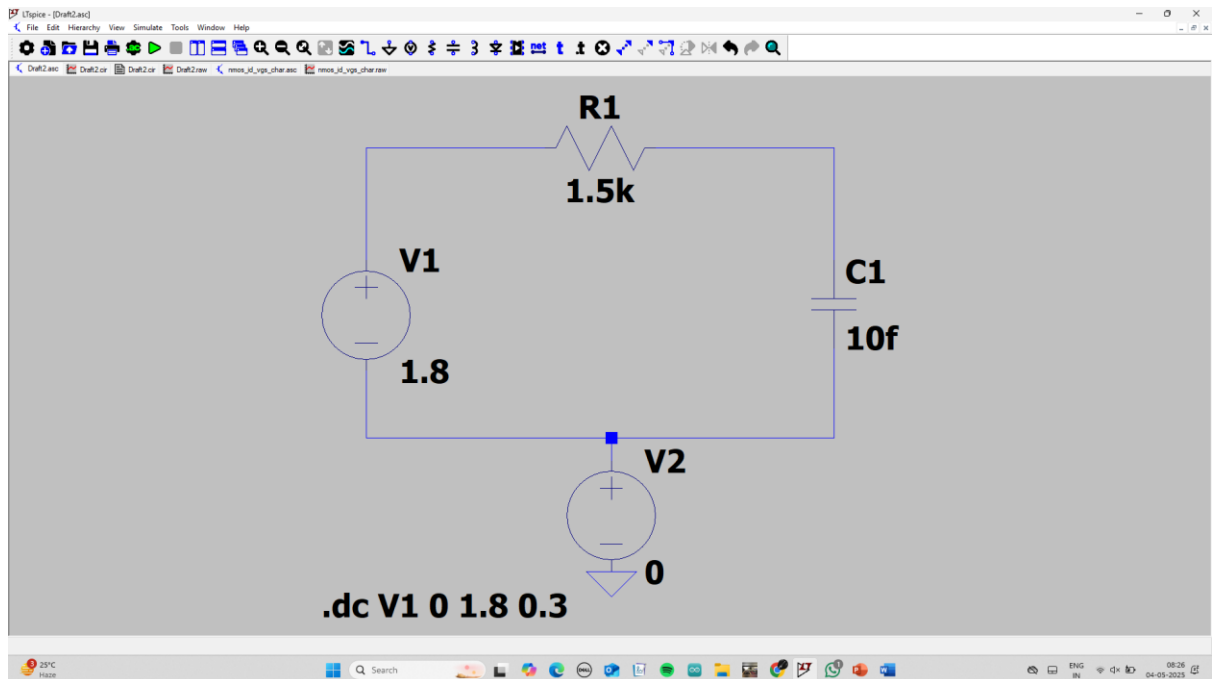


CIRCUIT SIMULATION IN LTspice

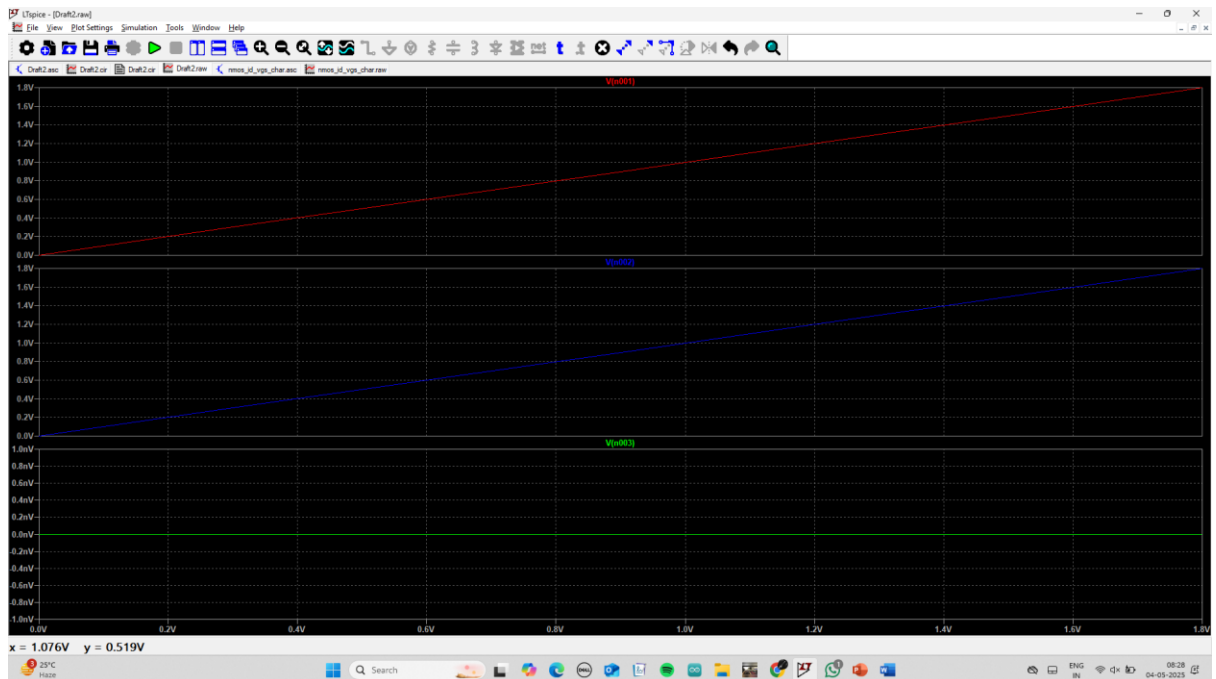
- By Er. Sailesh Mishra

1. Make the circuit :



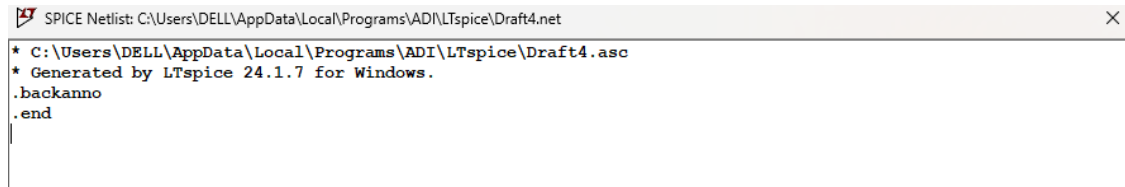
.dc V1 0 1.8 0.3 with this DC command & Run

2. Output graph showing V1 = 1.8V , I(R) & I(C) :



PARAMETRIC ANALYSIS

1. **Ctrl + N** (New schematic editor opens) -> **view -> SPICE netlist**

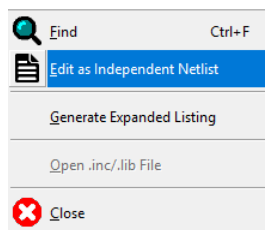


The screenshot shows a dialog box titled "SPICE Netlist: C:\Users\DELL\AppData\Local\Programs\ADI\LTspice\Draft4.net". The text inside the dialog box is as follows:

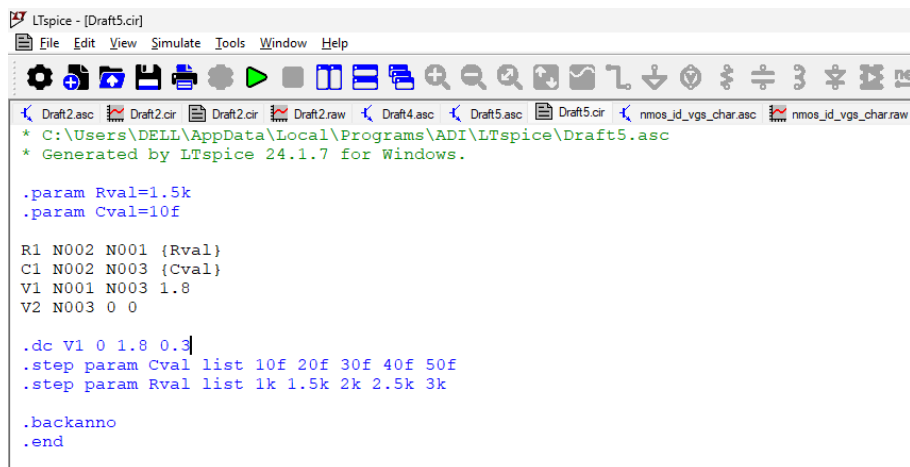
```
* C:\Users\DELL\AppData\Local\Programs\ADI\LTspice\Draft4.asc  
* Generated by LTspice 24.1.7 for Windows.  
.backanno  
.end
```

This kind of dialog box will open

2. **Right click on the white background -> Edit as Independent Netlist**



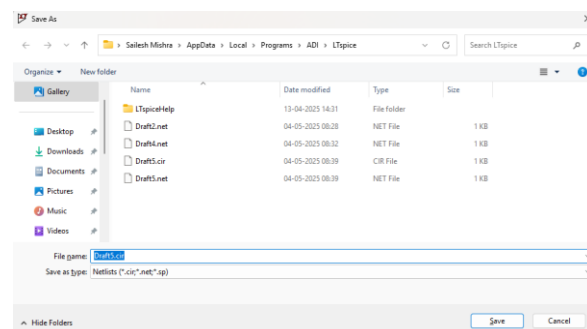
3. **Write the netlist manually (our netlist is Draft5.cir) :**



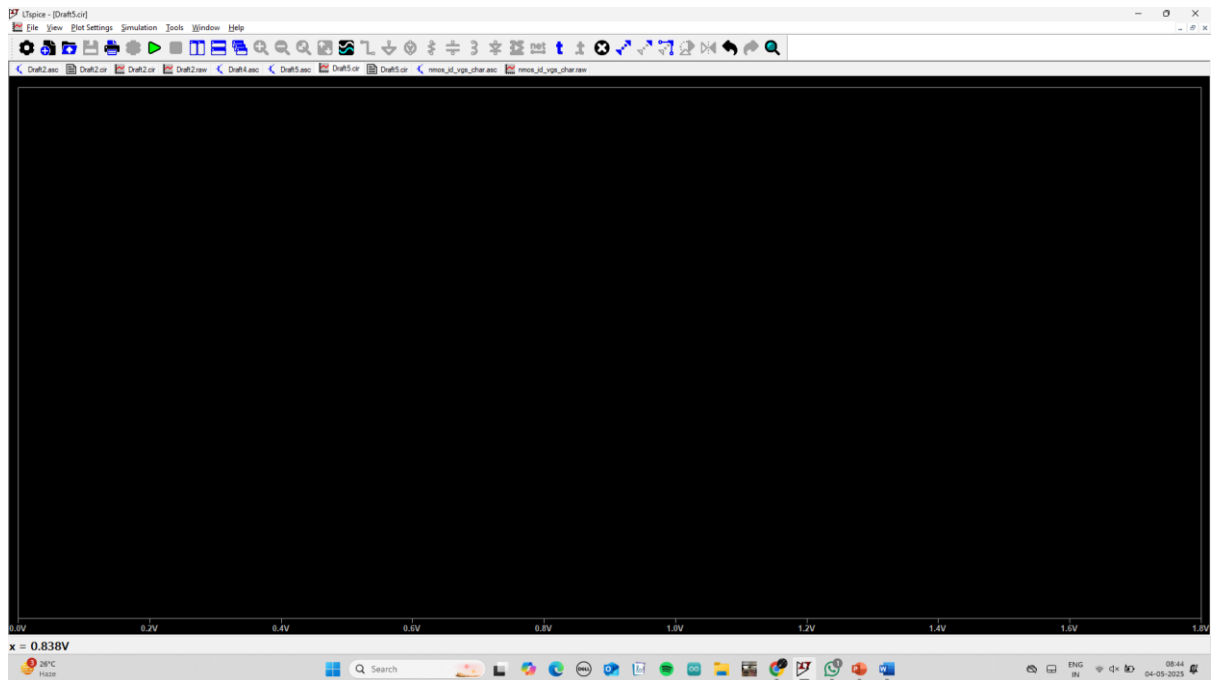
The screenshot shows the LTspice schematic editor with the netlist for Draft5.cir. The netlist content is as follows:

```
* C:\Users\DELL\AppData\Local\Programs\ADI\LTspice\Draft5.asc  
* Generated by LTspice 24.1.7 for Windows.  
  
.param Rval=1.5k  
.param Cval=10f  
  
R1 N002 N001 {Rval}  
C1 N002 N003 {Cval}  
V1 N001 N003 1.8  
V2 N003 0 0  
  
.dc V1 0 1.8 0.3  
.step param Cval list 10f 20f 30f 40f 50f  
.step param Rval list 1k 1.5k 2k 2.5k 3k  
  
.backanno  
.end
```

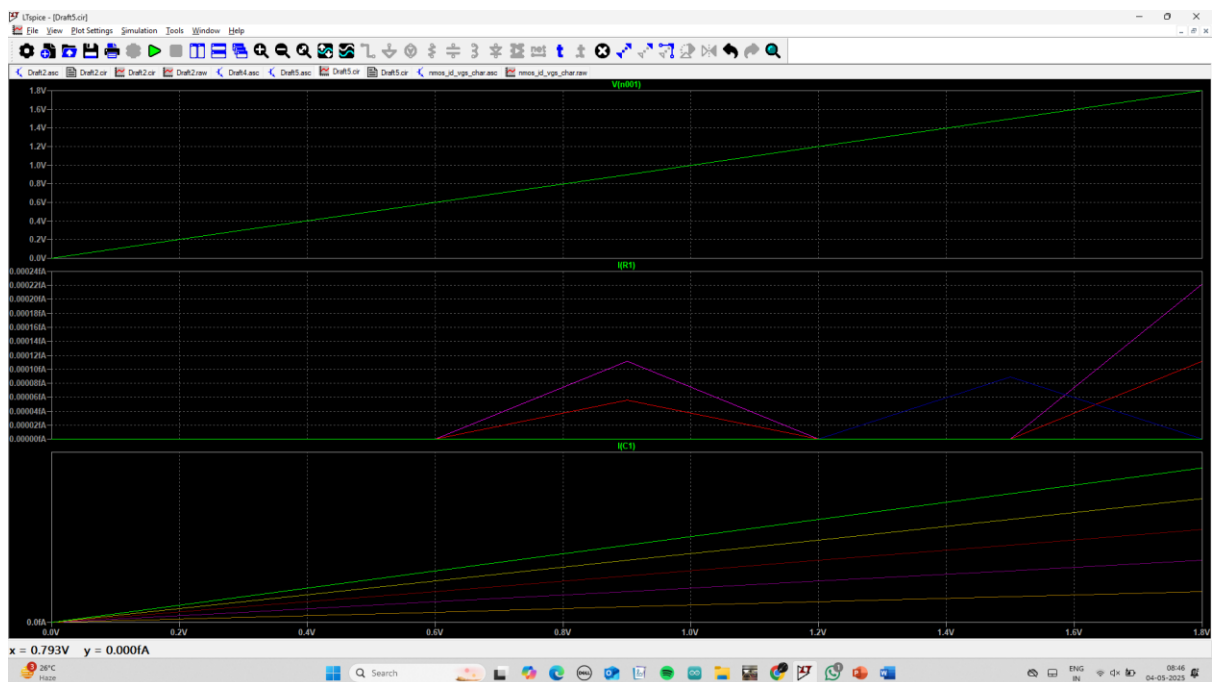
4. **File -> Save As -> Draft5.cir**



5. Then click on this  (Run button) for simulation (a black screen will appear) :



6. Right click -> add Traces -> plot the required signals/nets :



(VALID)

$$V_C(t) = V_{\text{supply}} \times (1 - e^{-t/\tau}) \rightarrow \text{transient Analysis}$$

Now, Let's understand the meaning of the NETLIST :

```
.param Rval=1.5k
.param Cval=10f
```

```
R1 N002 N001 {Rval}
C1 N002 N003 {Cval}
V1 N001 N003 1.8
V2 N003 0 0
```

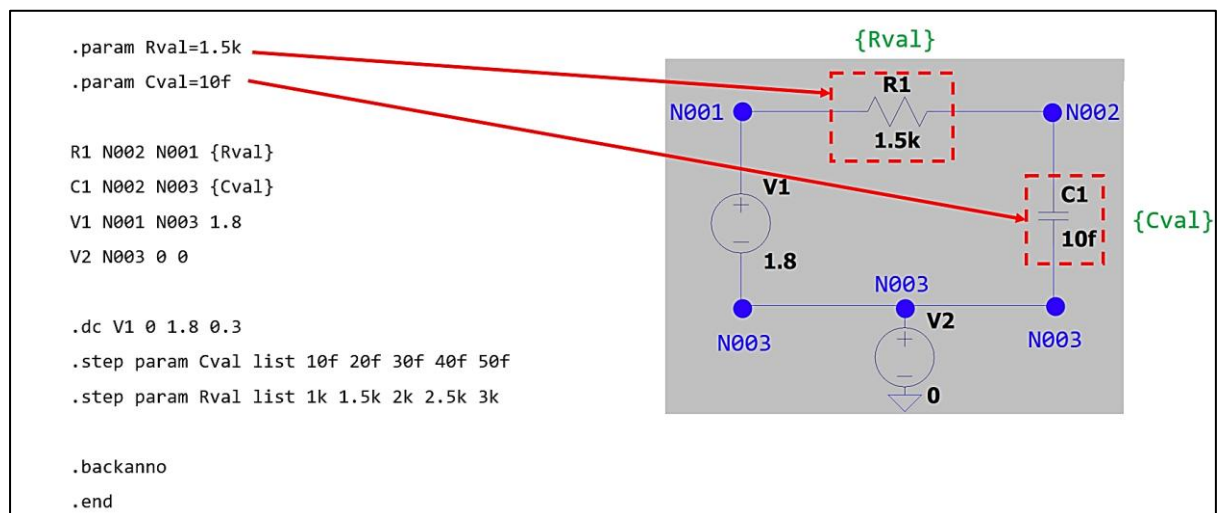
```
.dc V1 0 1.8 0.3
.step param Cval list 10f 20f 30f 40f 50f
.step param Rval list 1k 1.5k 2k 2.5k 3k
```

```
.backanno
.end
```

Mathematical
Textual
representation of
Electrical/Electronic
circuit(s).

CS $\rightarrow$ PI, PID, $\rightarrow$ I also

$$T.F(s) = \frac{N(s)}{D(s)} \rightarrow \checkmark \rightarrow \bigcirc$$



.param Rval=1.5k

- **.param** $\rightarrow$ Declares a parameter (user-defined variable).
- **Rval** $\rightarrow$ The name of the resistor value parameter.
- **=1.5k** $\rightarrow$ Sets the default value of Rval to 1.5 kilo-ohms (1500 ohms).

.param Cval=10f

- **Cval** $\rightarrow$ Name of the capacitor value parameter.
- **=10f** $\rightarrow$ Sets the default value of Cval to 10 femtofarads (10×10^{-15} F).

R1 N002 N001 {Rval}

- **R1** → Resistor name.
- **N002 and N001** → Nodes between which the resistor is connected.
- **{Rval}** → The value of the resistor is substituted from the parameter Rval.

C1 N002 N003 {Cval}

- **C1** → Capacitor name.
- **N002 and N003** → Nodes for the capacitor terminals.
- **{Cval}** → Uses the parameter Cval as the capacitor value.

V1 N001 N003 1.8

- **V1** → Voltage source name.
- **N001 and N003** → Positive and negative terminals respectively.
- **1.8** → Constant voltage of 1.8V applied from N001 to N003.

V2 N003 0 0

- **V2** → Another voltage source (used to anchor node N003 to ground).
- **N003 and 0** → Node N003 is connected to ground.
- **0** → Voltage = 0V (acts as a DC ground tie).

.dc V1 0 1.8 0.3

- **.dc** → Tells LTspice to perform a DC sweep analysis.
- **V1** → Sweeps the voltage source V1.
- **0 1.8 0.3** → Start at 0V, stop at 1.8V, step by 0.3V.

.step param Cval list 10f 20f 30f 40f 50f

- **.step** → Runs a parametric sweep.
- **param Cval** → Sweeps the parameter Cval.
- **list 10f 20f 30f 40f 50f** → Tries 5 different values for Cval (from 10 femtofarads to 50 femtofarads).

.step param Rval list 1k 1.5k 2k 2.5k 3k

- **Sweeps Rval** through 5 values: 1kΩ to 3kΩ.

Because this comes **after** the .step Cval, LTspice interprets this as **nested sweep** (for each Rval, it loops through all Cval values).

.backanno

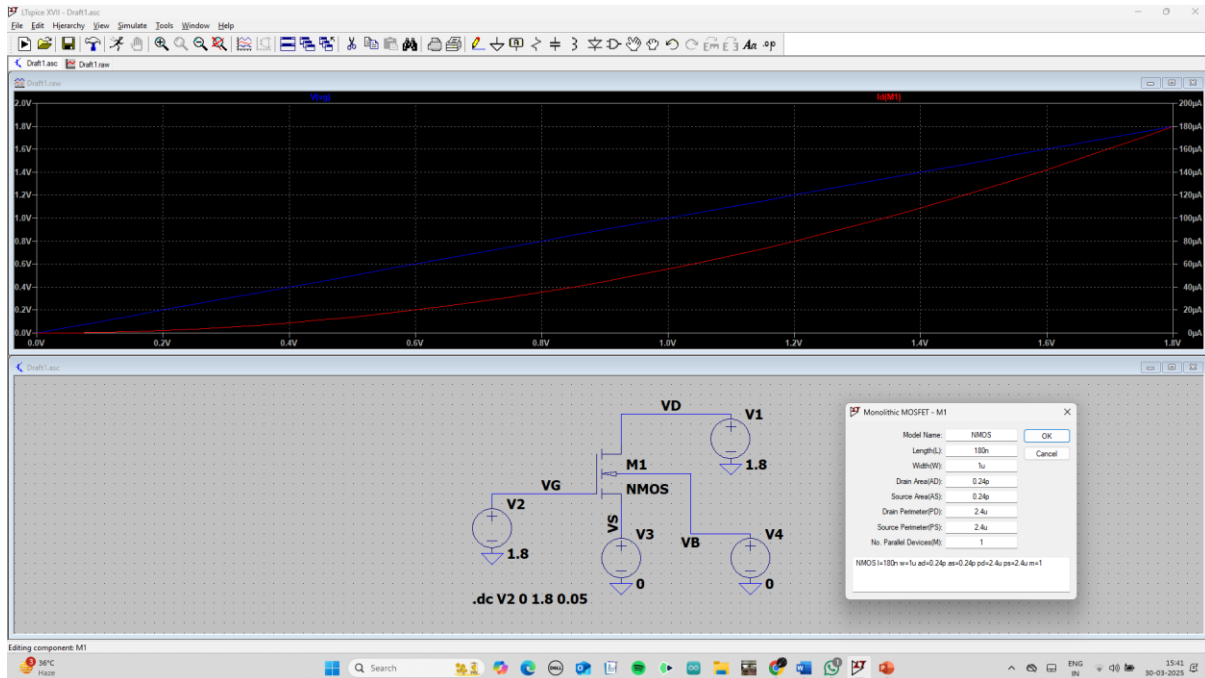
- **Optional** directive used for back-annotation of node names or properties from schematic (used in GUI version).

.end

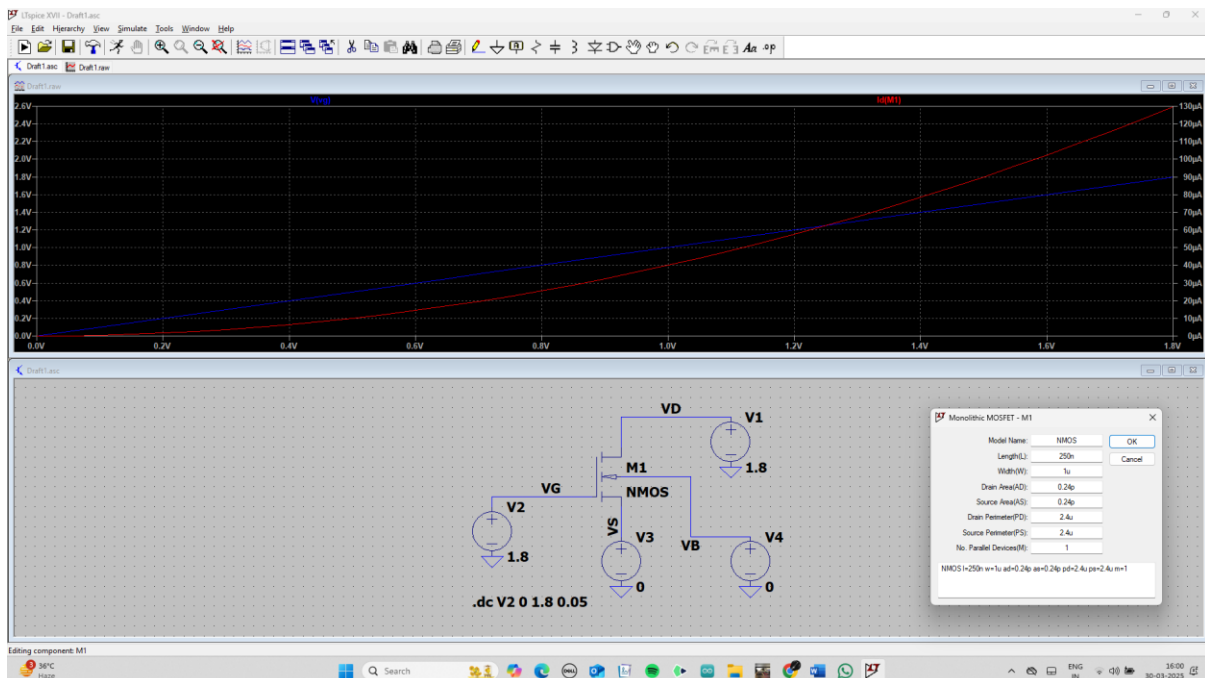
- **Ends the netlist.**

MOSFET CHARACTERIZATION

ID vs VGS

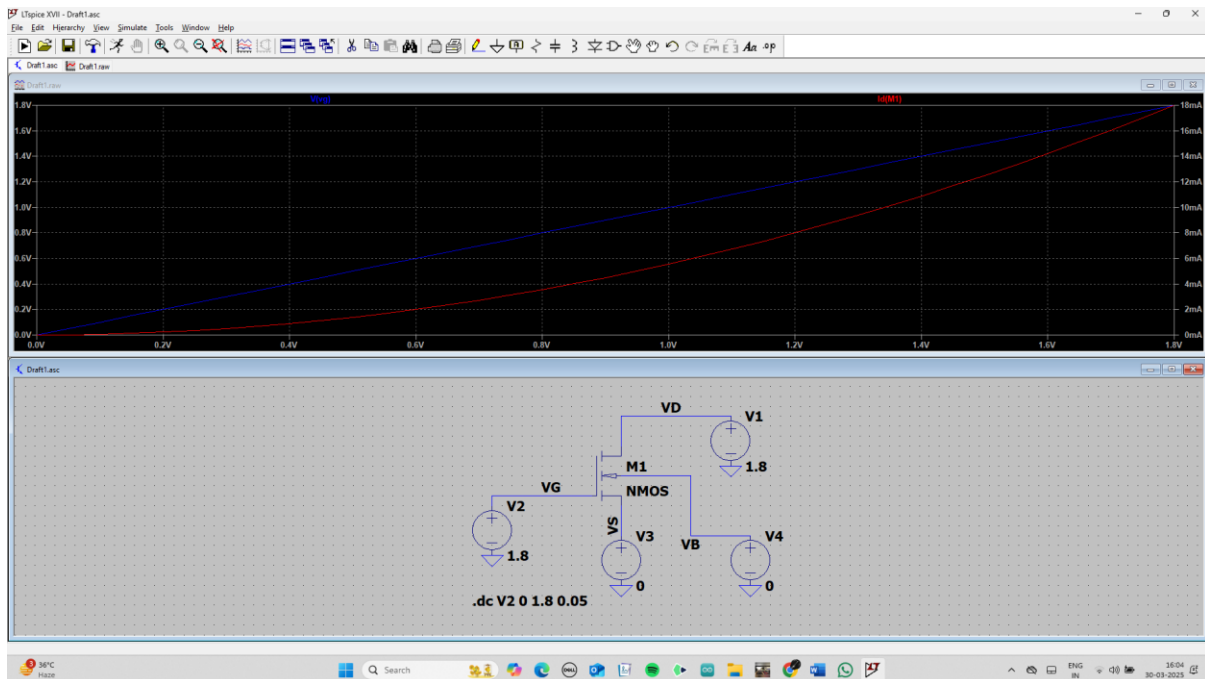


ID vs VGS with the following paramters of $L = 180\text{nm} = 0.18\mu\text{m}$ & $W = 1\mu\text{m}$, ID is $180\mu\text{A}$.

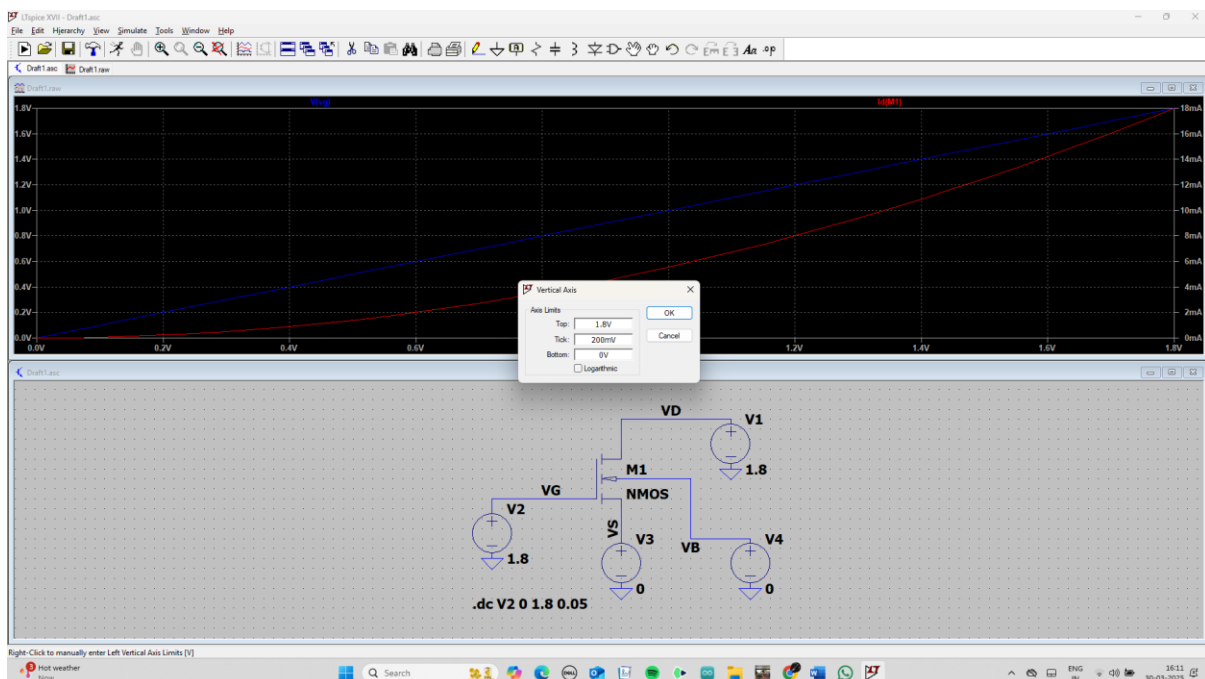


Everything is constant , Only L is increased to $250\text{nm} = 0.25\mu\text{m}$, Now see the ID vs V_{GS} , ID decreased to $90\mu\text{A}$.

Now , Let's increase the W from 1um to 100um keeping L = 180nm as constant and let's see the change , theoretically ID should increase proportionally .

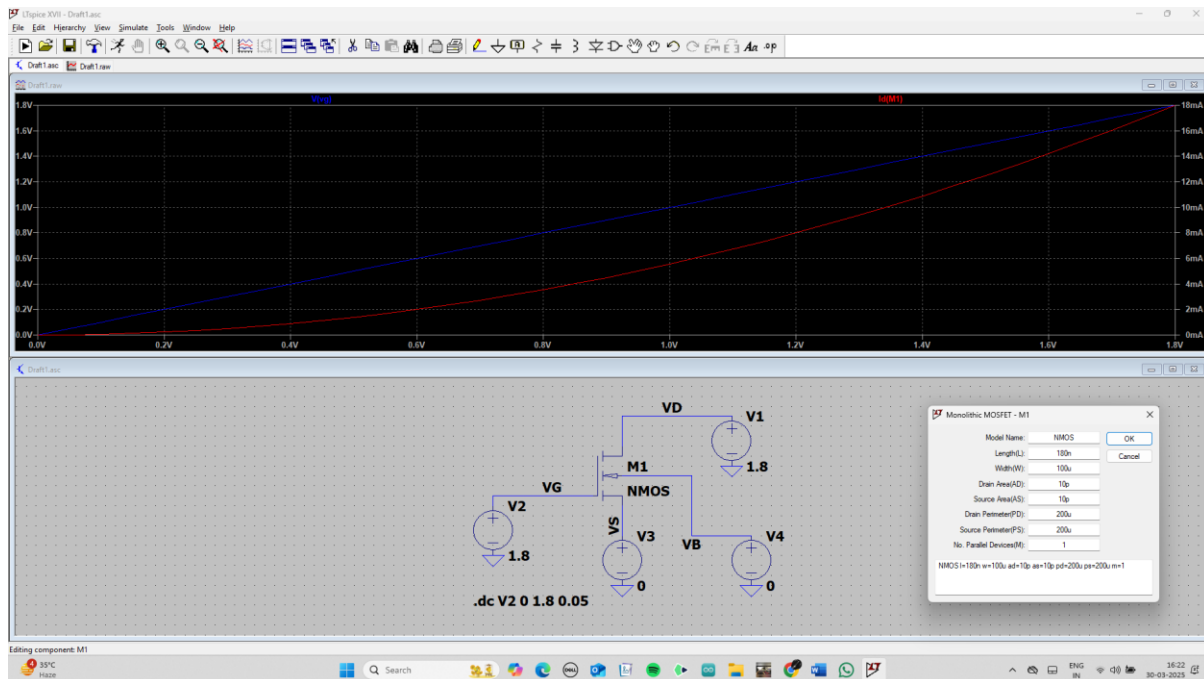


ID increased to 18mA



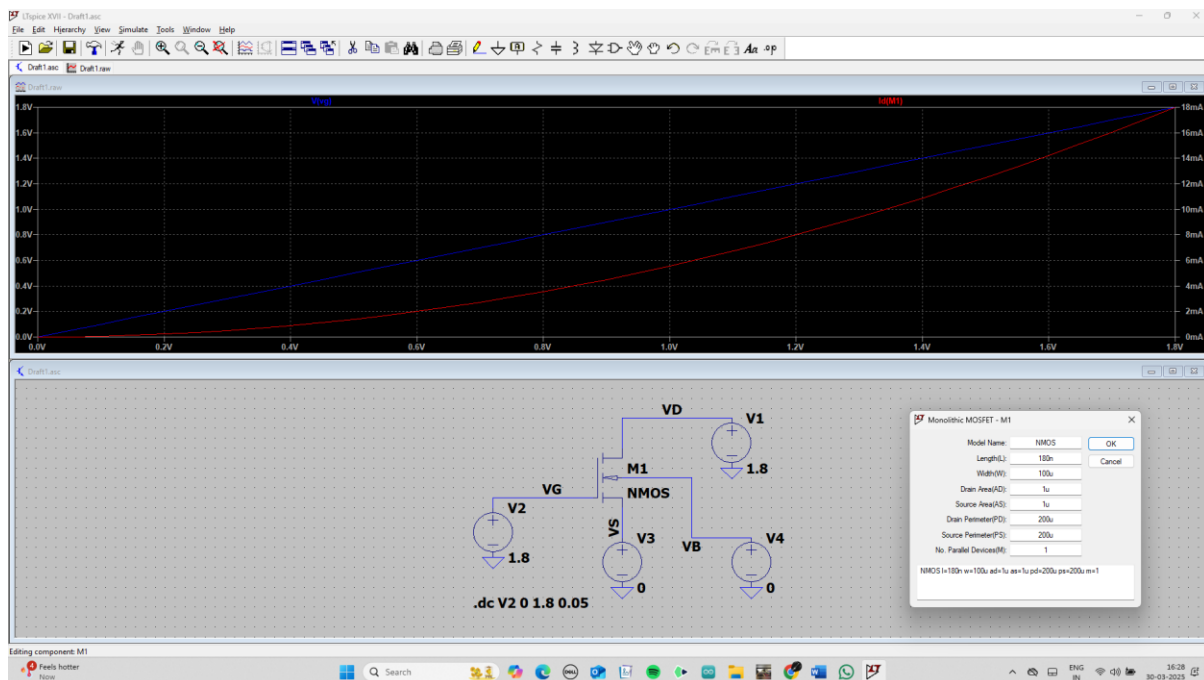
If we will Right click on the y-axis , and check the logarithmic scale , we will get **logID vs VGS** graph , Henceforth we can analyse the MOSFET's operation in SUB-THRESHOLD REGION !! that thing we will do it later on actually

Now let's see with the same specifications of MOSFET , and keeping everything constant
Let's change the S & D perimeter and Area and see what happens let's evaluate . . .

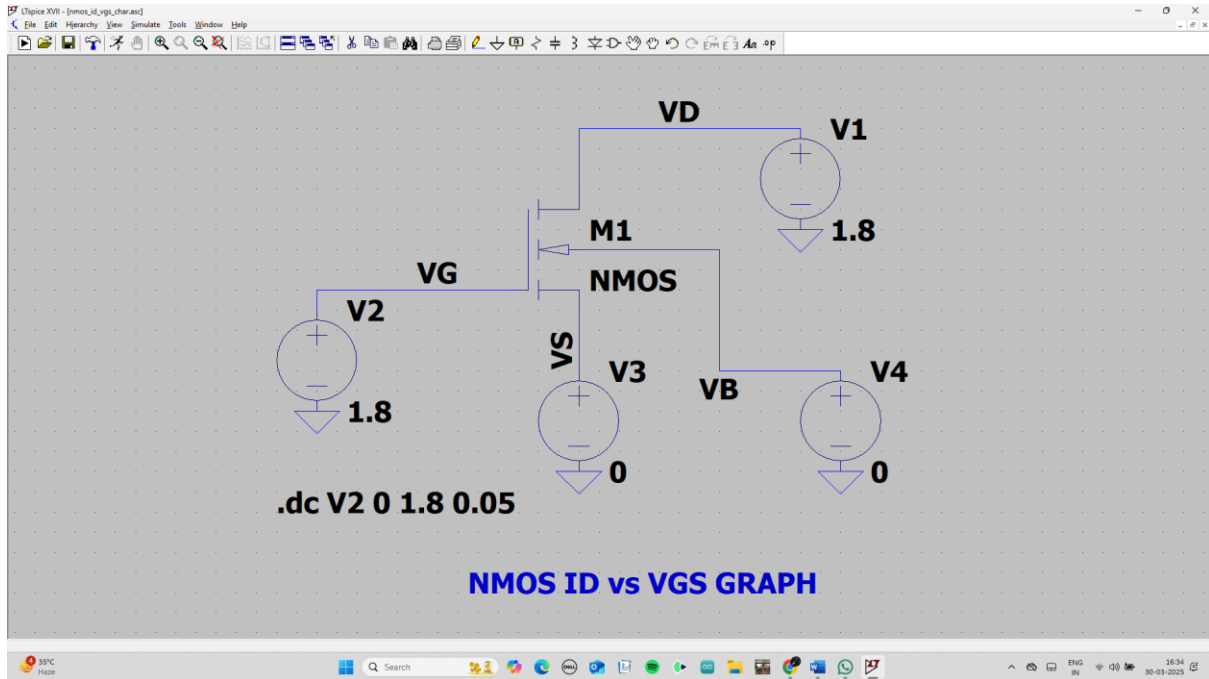


So , NO !! “ *keeping the W/L constant i.e., 100u/0.18um and increasing the AD , AS , PD , PS doesn't modify (increase or decrease) the ID in DC ANALYSIS !* ”

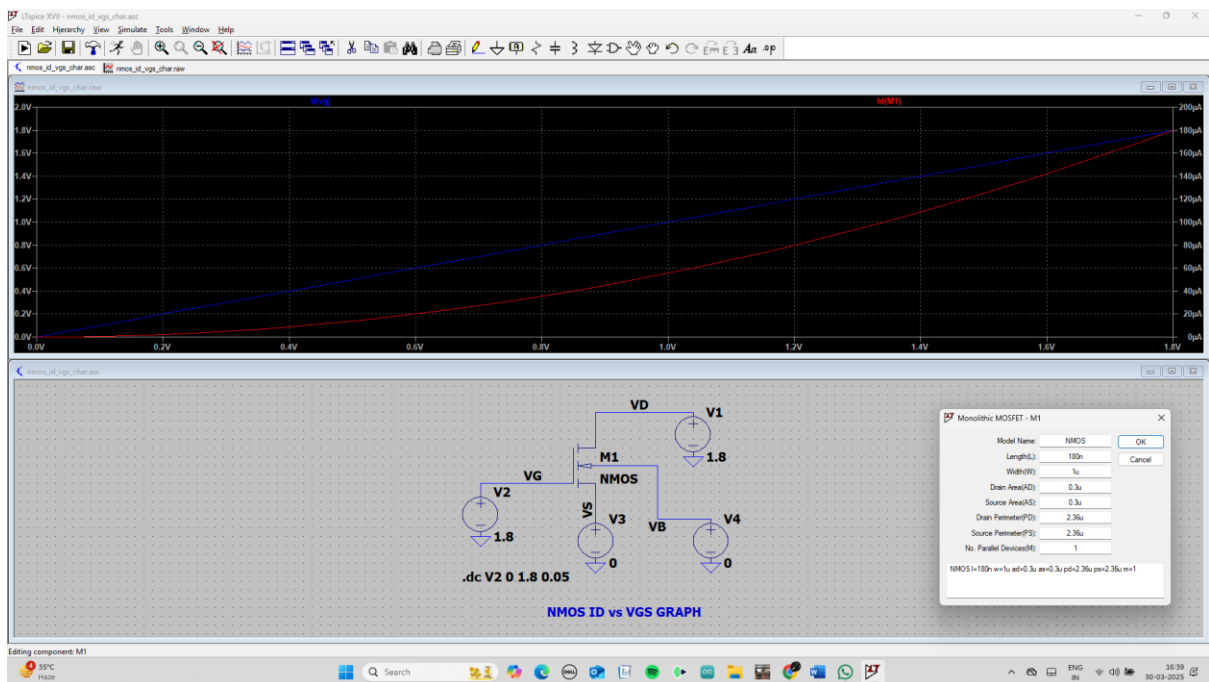
So , No matter how much you increase or decrease the AS , AD , PS , PD , ID will not change at all in DC ANALYSIS , But in AC ANALYSIS & TRANSIENT ANALYSIS , ID will change due to parasitic resistances and junction capacitances !!



In the above , simulation I have increased the AS , AD , PS , PD but ID remained the same i.e., 18mA ! But remember this is DC ANALYSIS !!



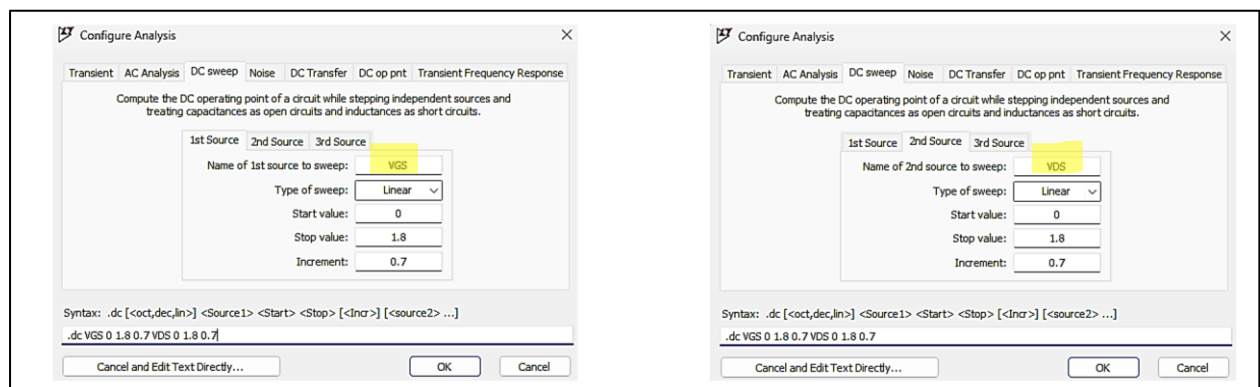
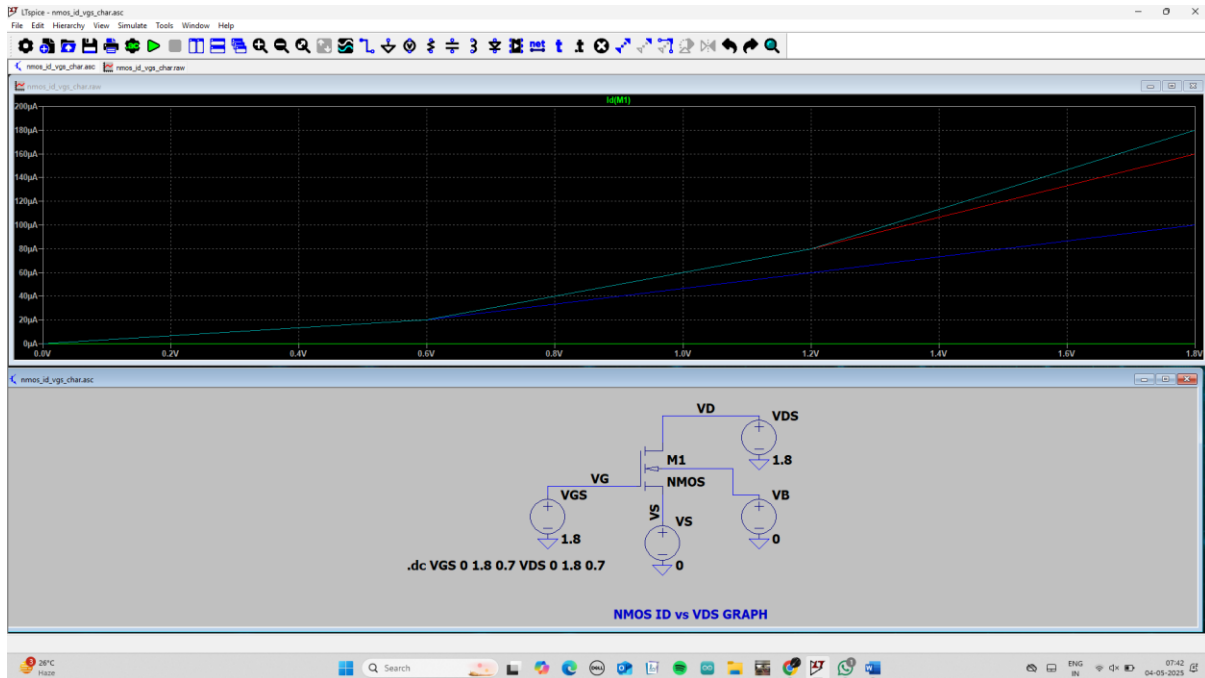
So , Throughout the analysis we will take an optimum W/L ratio with PS , PD , AS , AD as this :



We will get $ID = 180\mu A$ that is standard !

Actually it's V_T is very very small, that's why I_D is getting much more at $V_G = 1.8V$

PARAMETRIC ANALYSIS

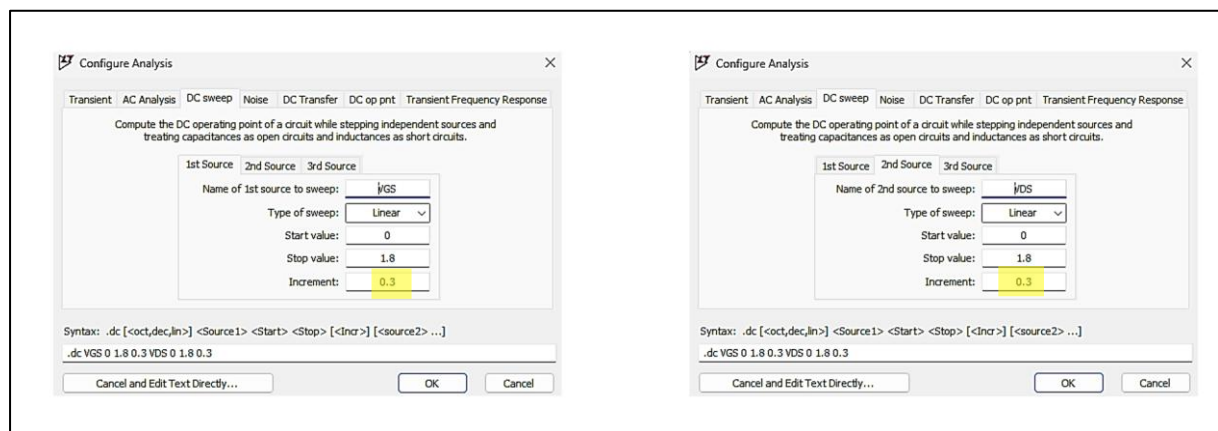
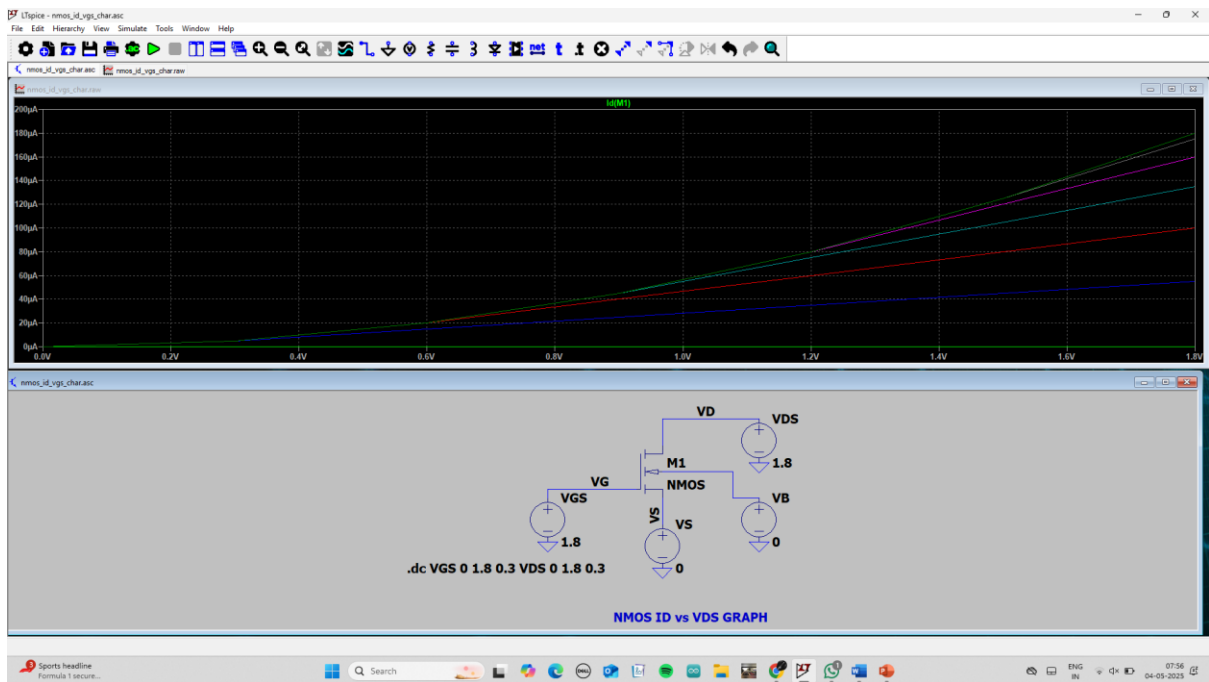


2 Commands :

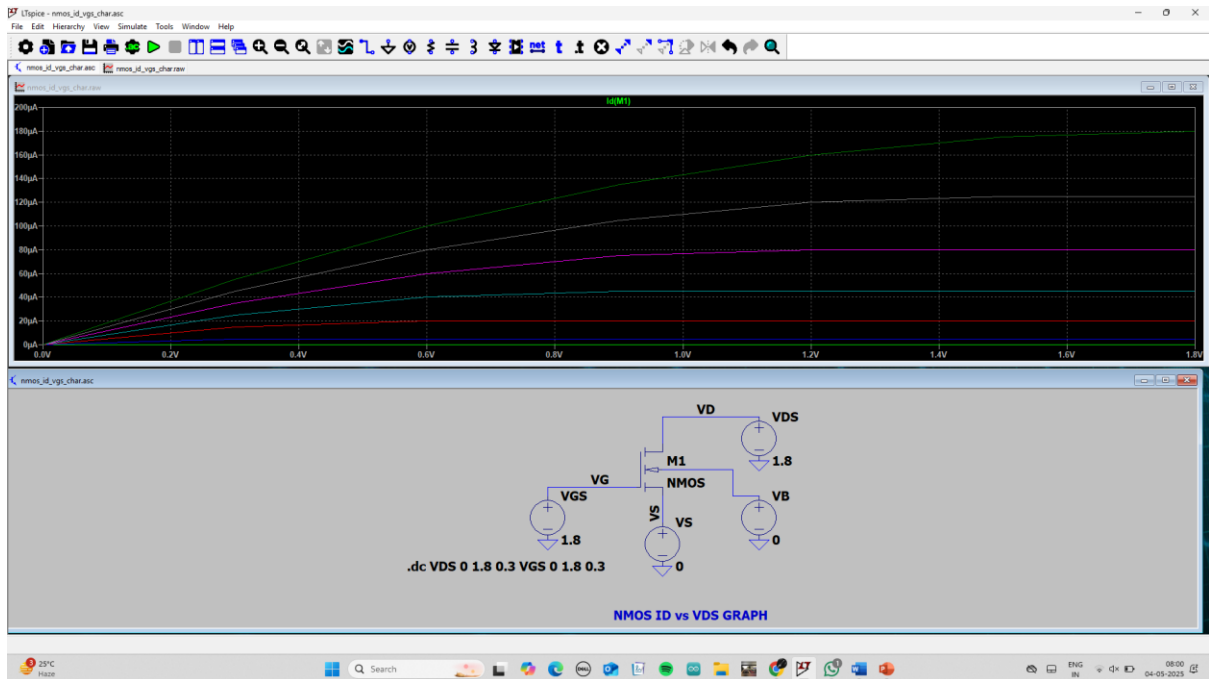
1. **.dc VGS 0 1.8 0.05** (I_D vs V_{GS} , $V_{DS} \rightarrow$ constant 1.8V :: only one curve)
2. **.dc VGS 0 1.8 0.7 VDS 0 1.8 0.7** (I_D vs V_{GS} , V_{GS} & V_{DS} both varying, see above screenshot :: V_{GS} = varying from 0 $\rightarrow$ 1.8 with a interval of 0.7V i.e @0V @0.7V @1.4V plus V_{DS} = varying from 0 $\rightarrow$ 1.8 with a interval of 0.7V i.e @0V @0.7V @1.4V)

If I want to have more curves, then I will decrease the interval from **0.7 to 0.3**, so I will get the graph at $V_{GS} = @0V @0.3V @0.6V @0.9V @1.2V @1.5V @1.8V$, Similarly I will decrease the interval from **0.7 to 0.3** for V_{DS} , so I will get the graph at $V_{DS} = @0V @0.3V @0.6V @0.9V @1.2V @1.5V @1.8V$

Let's see it in the graph now . . .

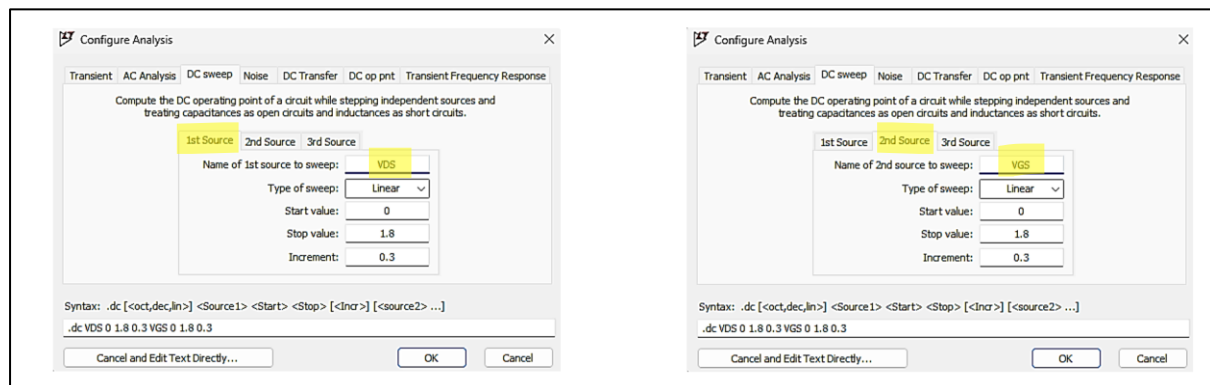


Now Let's see the ID vs VDS curve ,



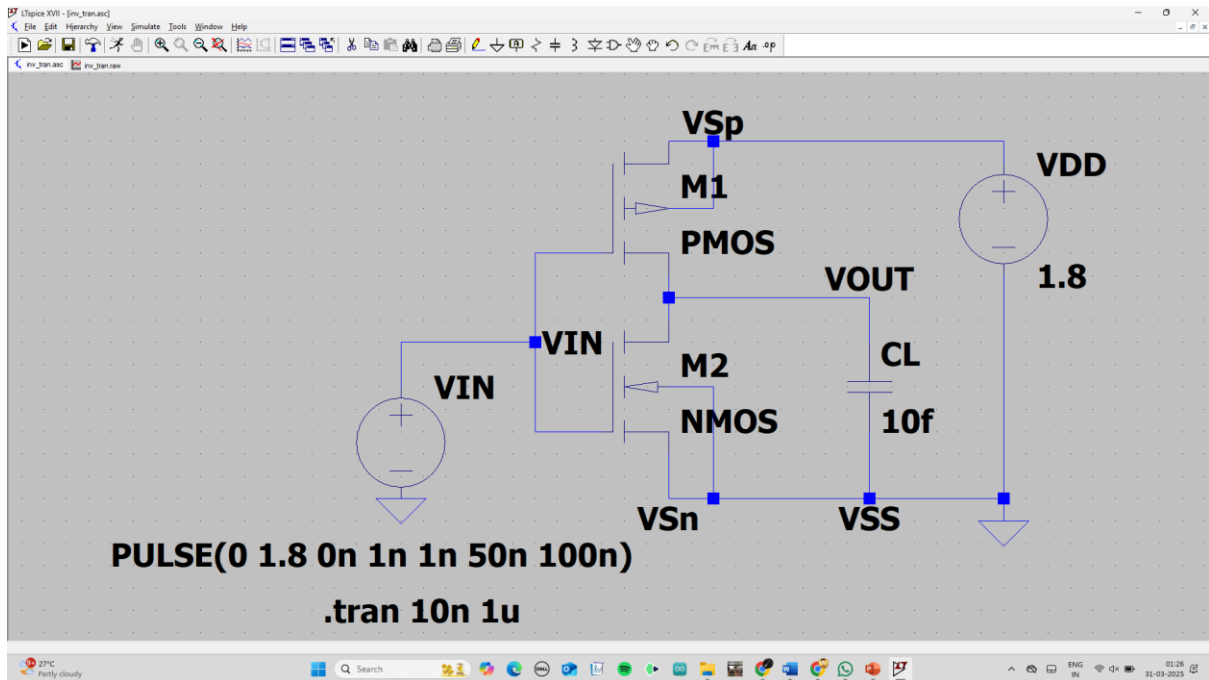
So , What we did ??

We just interchanged the 1st source to V_{DS} and 2nd source to V_{GS} , that's it !!

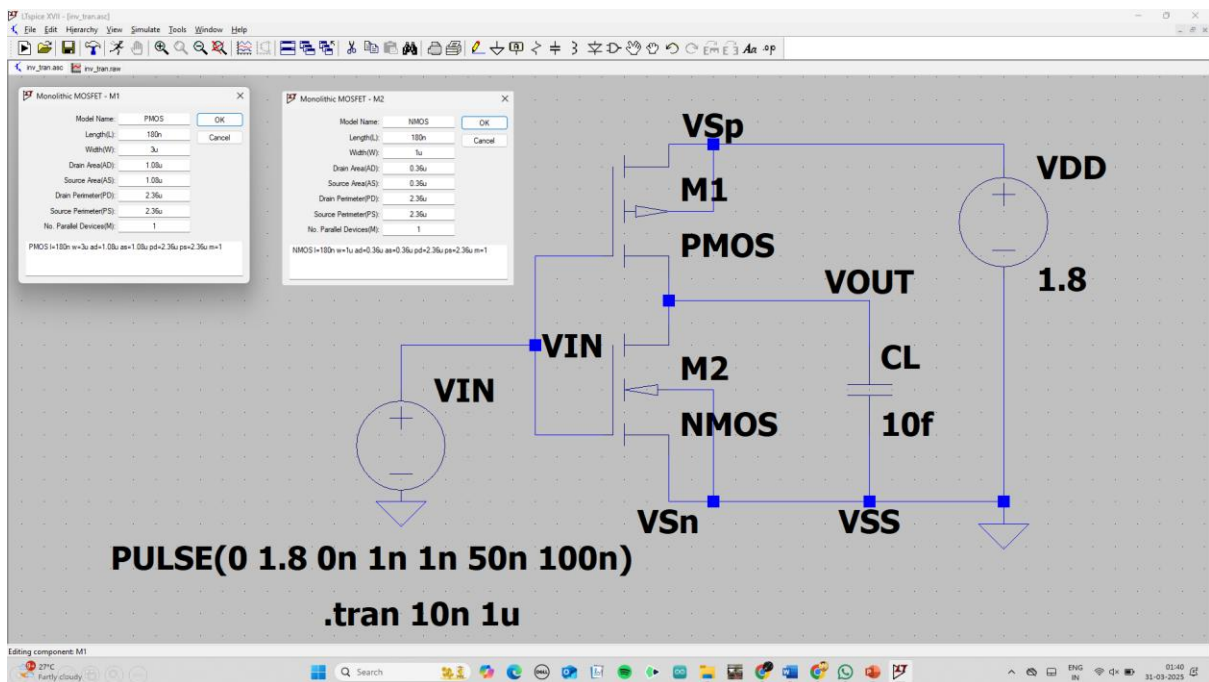


So , Let's do the AC analysis or TRANSIENT Analysis . . .

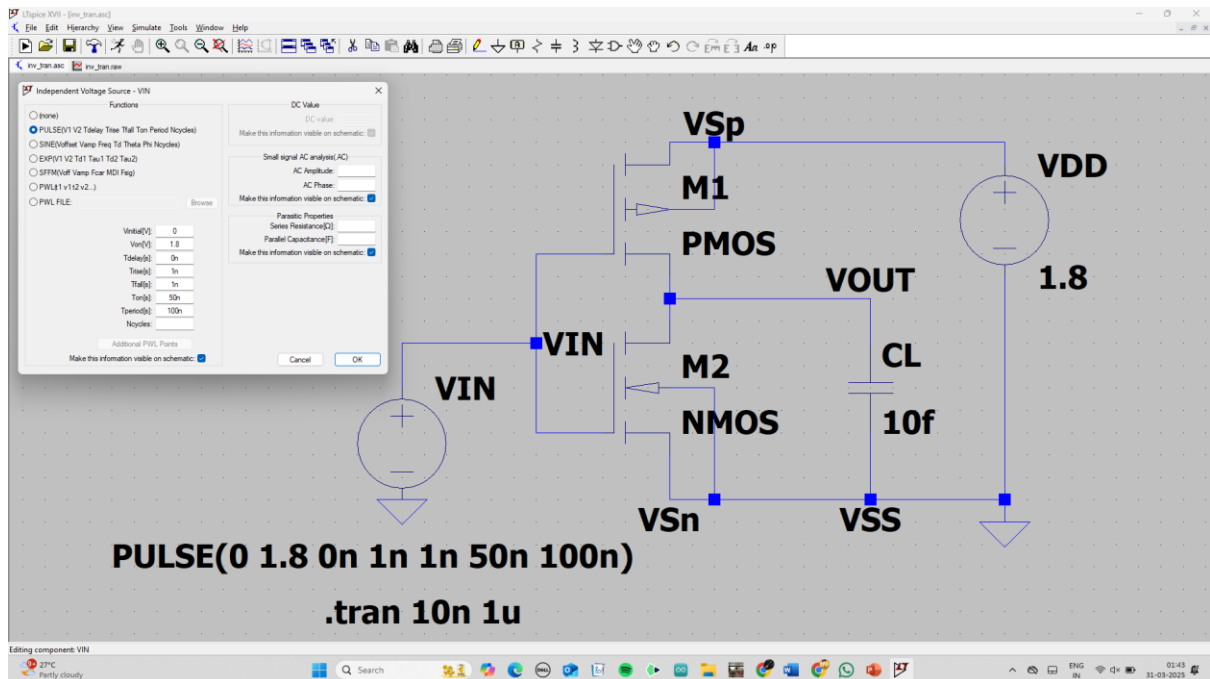
CMOS INVERTER



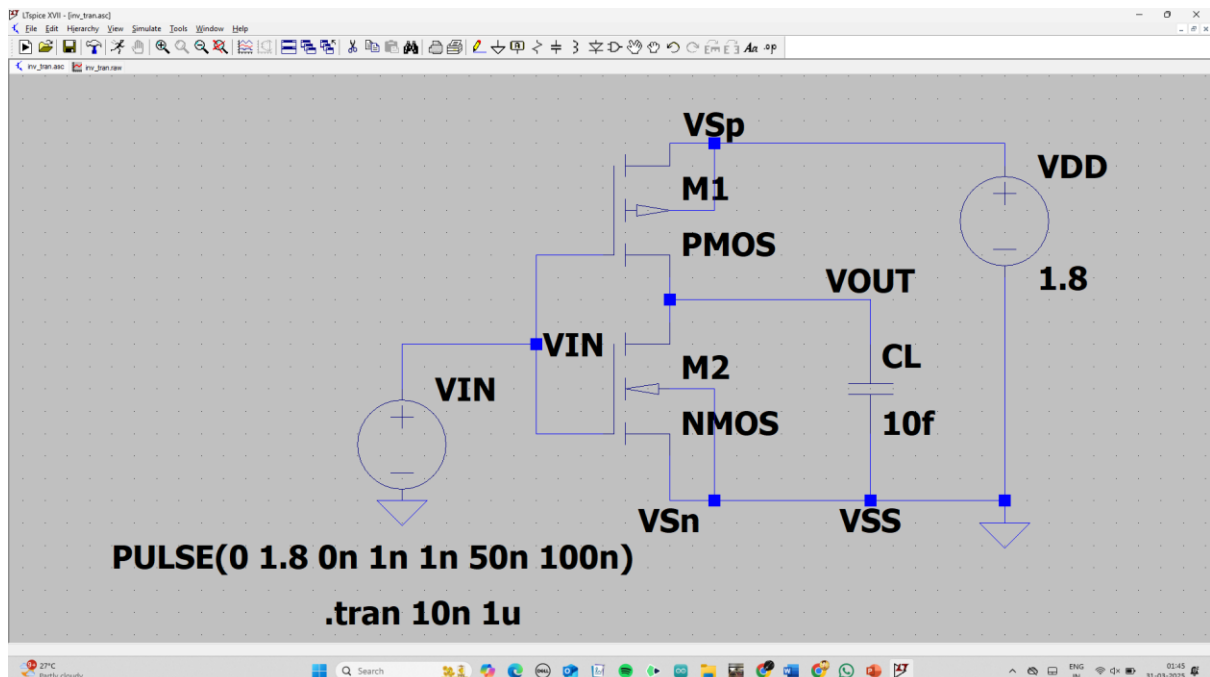
1. Get pmos4 & nmos4 from components pane (fn+F2) .
2. Parameterize them with W, L, AS , AD , PS , PD & M by Right-clicking on PMOS & NMOS.



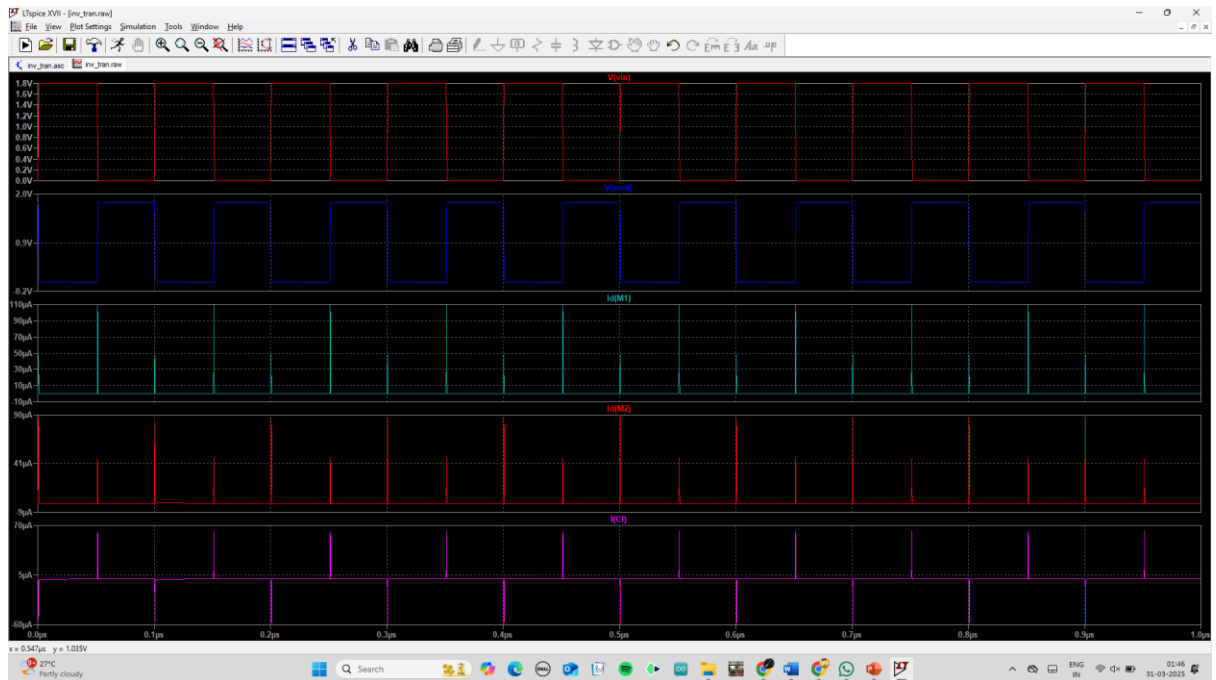
- Again , go to the component's pane and bring voltage , Right click then advanced PULSE , then paste this : for VIN or VGS pulsating signal '0' '1' '0' '1' '0' '1' '0' '1'.



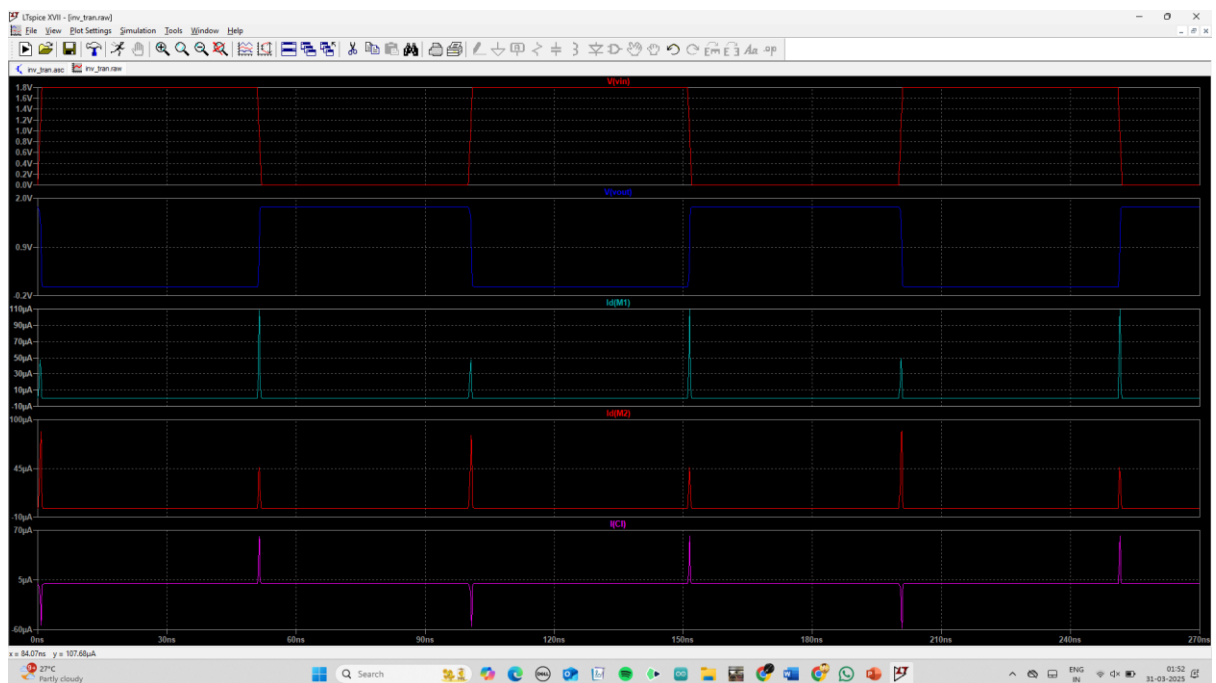
- Name the labels (fn+F4) , Remember if you will give same name , during netlisting and simulation , tool will get confused whether the Source is of PMOS or NMOS , Hence mention in the label name itself , VSp for PMOS and VSnp for NMOS .
- Now, press Run button  (man running depiction at right of hammer sign) , The .op will appear , write there **.tran 10n 1u** .



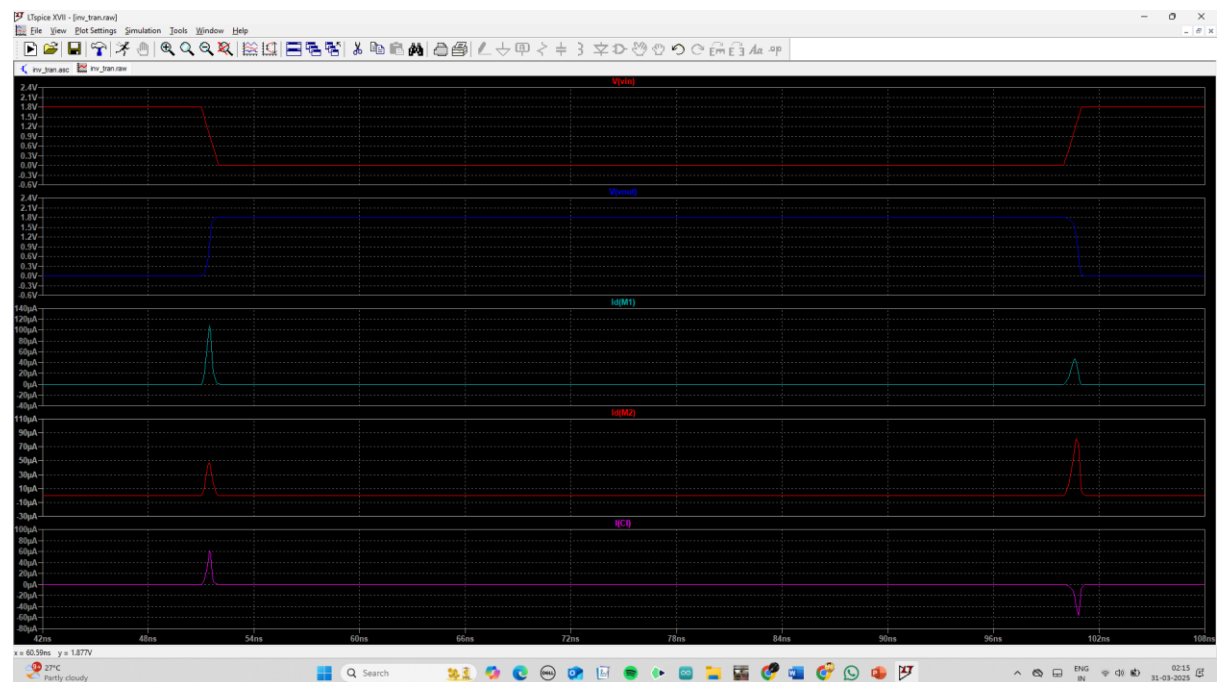
6. Now , Our CMOS inverter schematic is ready for TRANSIENT ANALYSIS .
7. Go to the waveform viewer of LTSpice , Right click on the black background and go to visible traces select the signals you want to analyse this is the best method otherwise click on the schematic and with probe touch the required terminals/ports to view the signals .



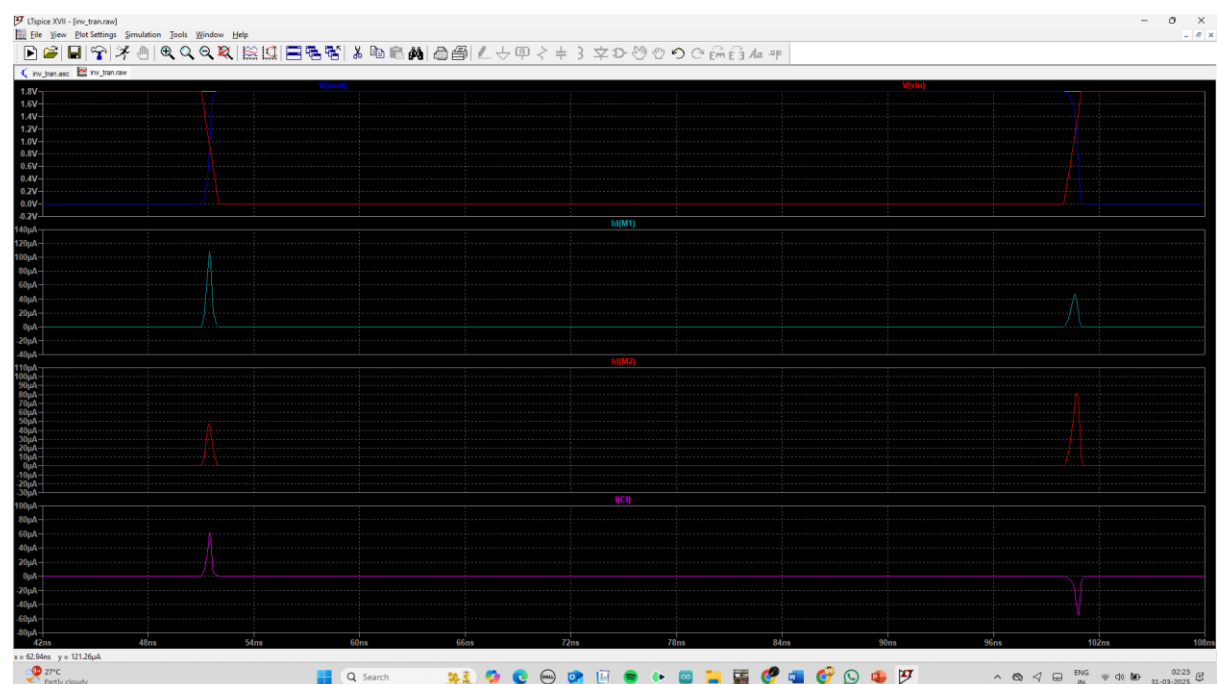
8. Now , to see any particular section of wave , pressing left mouse button drag how much section you want to analyse and accordingly do it , it will get zoomed .



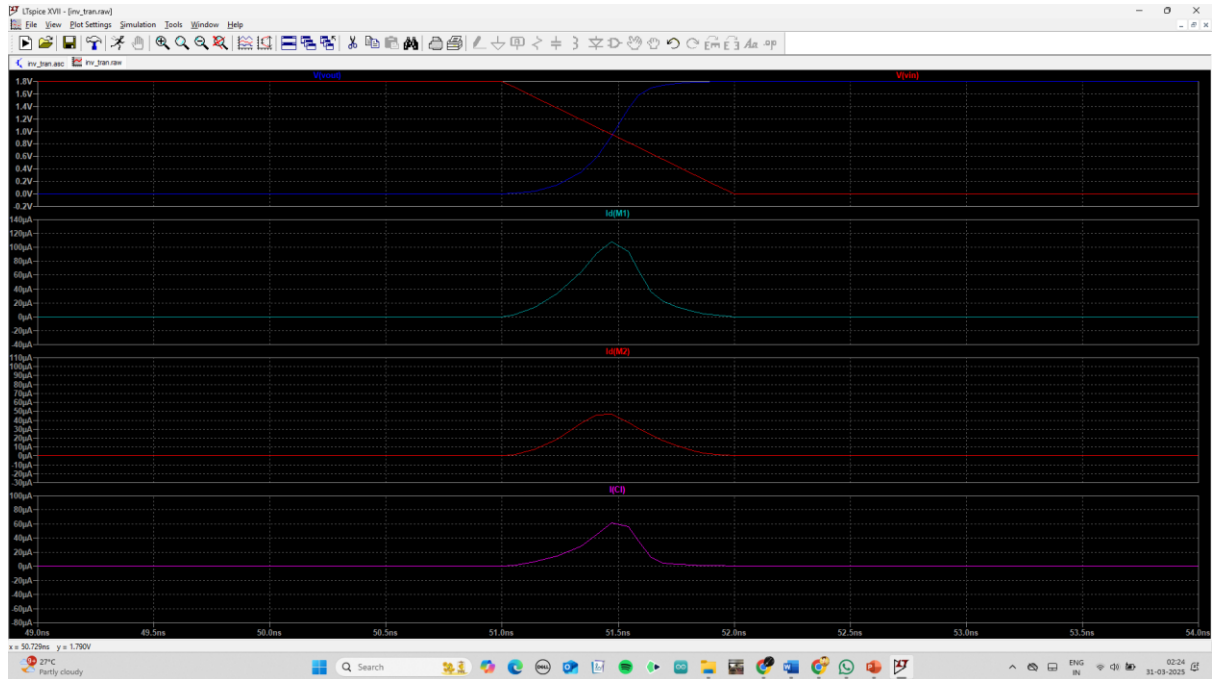
9. Further zoom to see the τ_{PLH} & τ_{PHL} , remember H $\rightarrow$ L of the output signal not the input signal similarly, L $\rightarrow$ H of the output signal not the input signal.



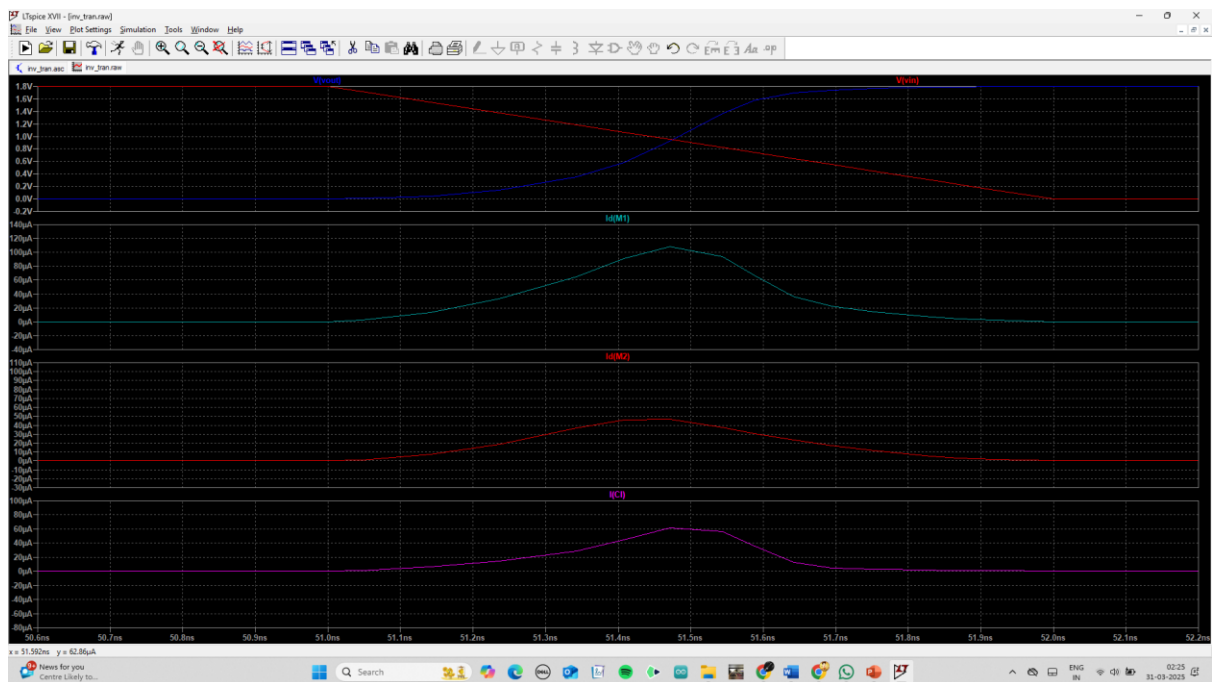
10. Now we will calculate the PD and we will also analyse whether changing (CASE 1 : INCREASING ; CASE 2 : DECREASING) the AS, AD, PS, PD , As in DC it doesn't matter , But in AC or TRANSIENT analysis , it will matter because of parasitic resistance and junction capacitance !! We will see whether the PD will INC. or DEC.
11. For this , We will merge both the vin and vout graphs and calculate the τ_{PLH} & τ_{PHL} .

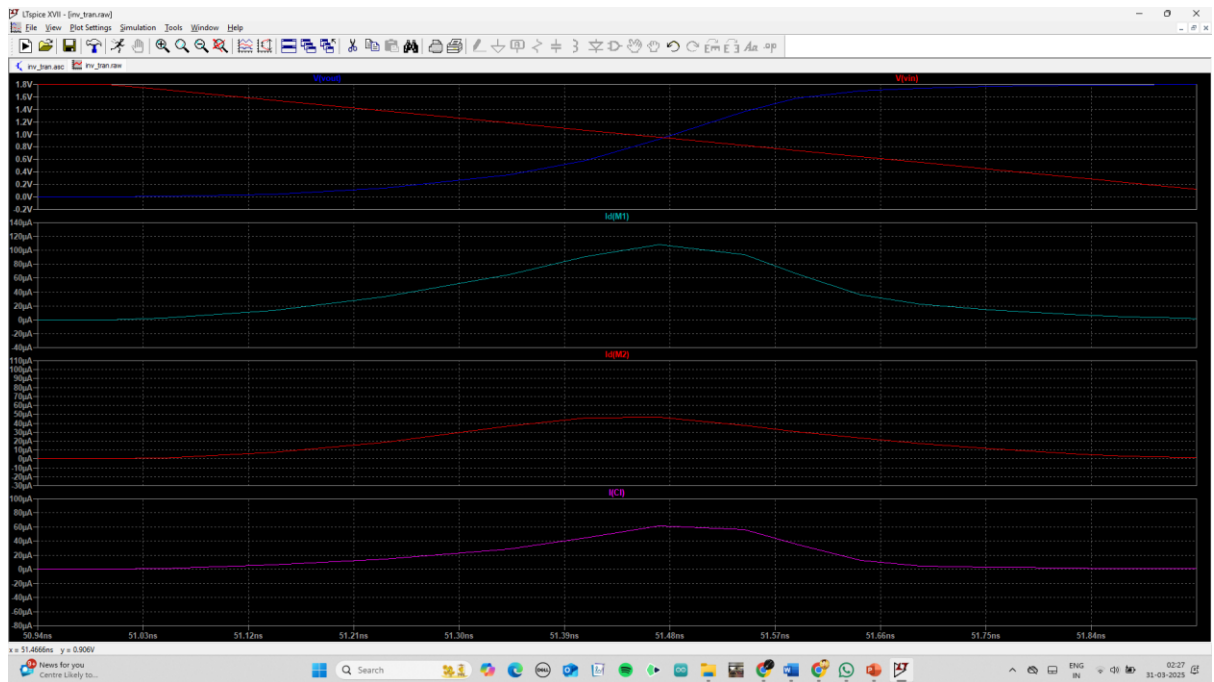


Again we will zoom . . .

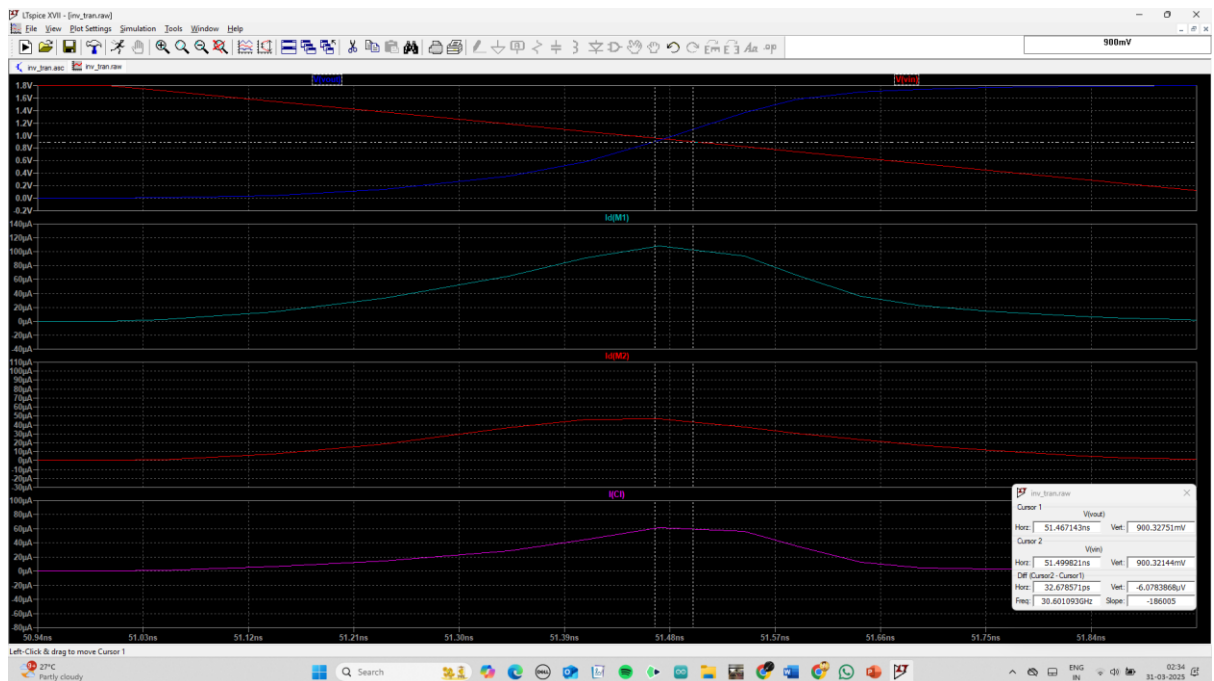


Further till we find 900mV i.e., the MID-VOLTAGE (1/2 OF THE VDD(supply_voltage)) : V50%

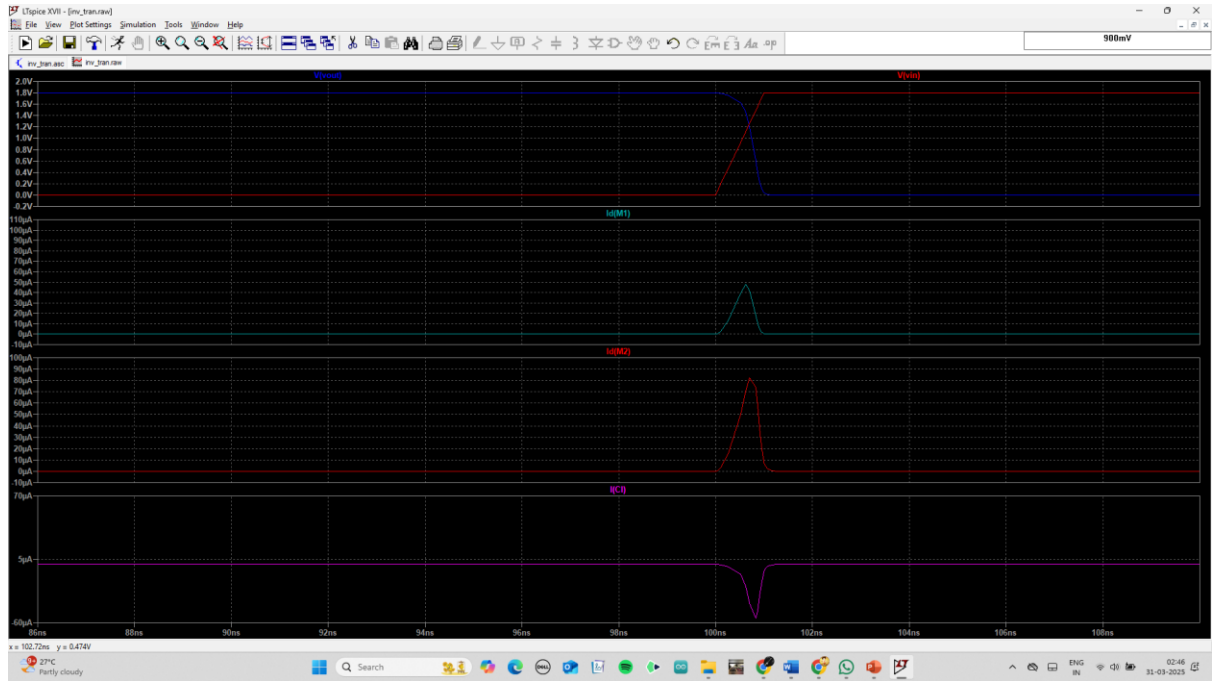




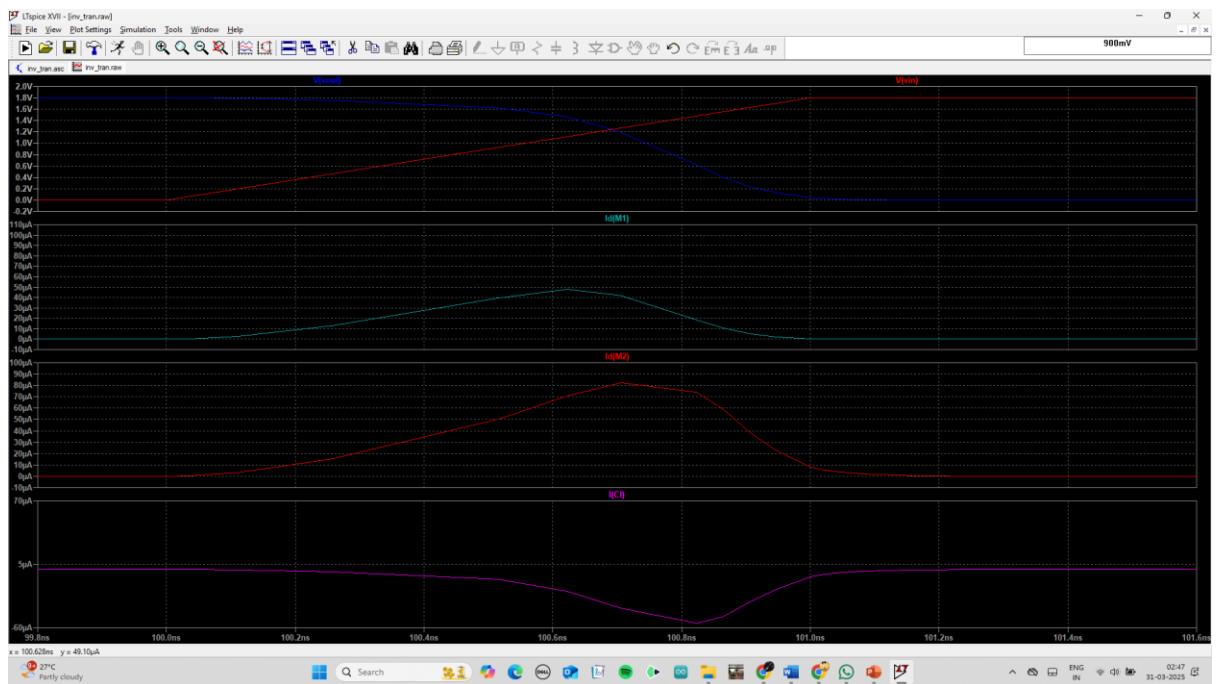
12. Click on the V(vin) , a yellow coloured point 2 will come corresponding to cursor 2 ; Click on the V(vout) , a yellow coloured point 1 will come corresponding to cursor 1 See the corresponding times (*in ns*) to Cursor 1 & Cursor 2 we will get τ_{PLH} i.e., $\text{RISE TIME} = 51.467143\text{ns} - 51.499821\text{ns} = -0.032678\text{ns} = 32.67857\text{ps}$, it's actually written in the small dialog box "*in picoseconds*" , Even the *Freq & Slope* is quoted that's very very low , HIGH SWITCHING SPEED !!

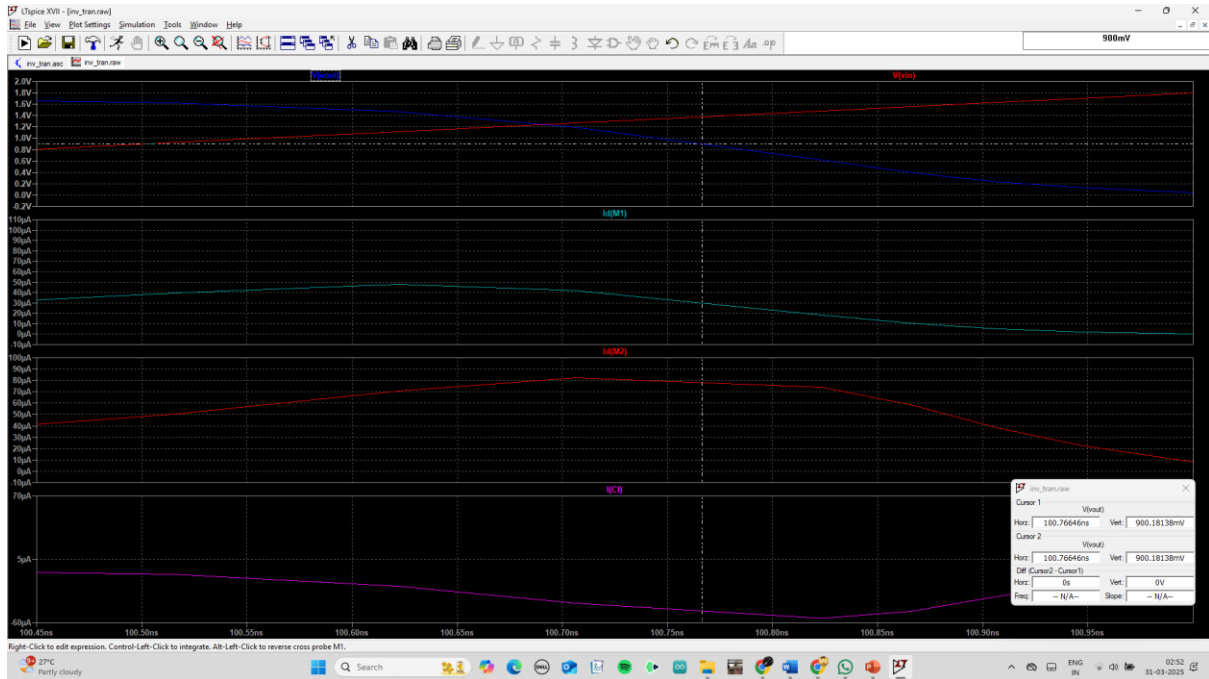


13. Similarly , Let's Calculate the τ_{PHL} i.e., FALL TIME ,



Further zoom . . .





14. Click on the V(vin) , a yellow coloured point 2 will come corresponding to cursor 2 ; Click on the V(vout) , a yellow coloured point 1 will come corresponding to cursor 1 See the corresponding times (*in ns*) to Cursor 1 & Cursor 2 , (*Actually the 2 points overlapped with each other that's why it's not showing 2 dotted lines , don't get confused*) ,we will get $T_{ao}(PHL)$ i.e., FALL TIME = $100.7664ns - 100.7664ns = 0ns$, that SIMPLY INSANE , HIGH SWITCHING SPEED !!

Generally , it's true because. . . NMOS has electrons whose mobility is FAST so it DISCHARGES very soon . . . the conduction is like LIT 🔥 🔥 🔥 .Hence the current rushing out too much .

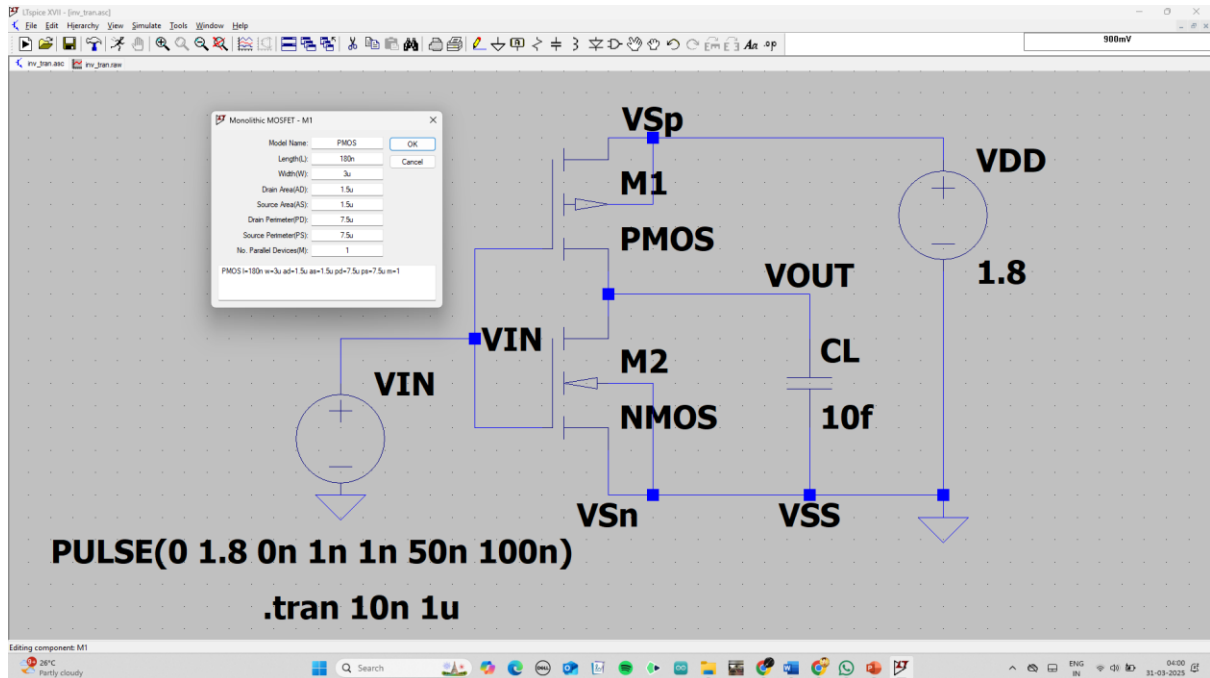
So , The **$P.D = T_{ao}(PLH) + T_{ao}(PHL) / 2 = RISE\ TIME + FALL\ TIME / 2 = 0.032678ns + 0ns / 2 = 0.016335ns = 16.335ps$** . Now Let's see if we increase the AREA OF 'S' & 'D' what happens . . . ???

2. Increased AS, AD, PS, PD (Higher Capacitance & Resistance → Expected to Increase Delay)			
Parameter	PMOS (M1)	NMOS (M2)	
AS	1.5 μm^2	0.5 μm^2	
AD	1.5 μm^2	0.5 μm^2	
PS	7.5 μm	3.0 μm	
PD	7.5 μm	3.0 μm	

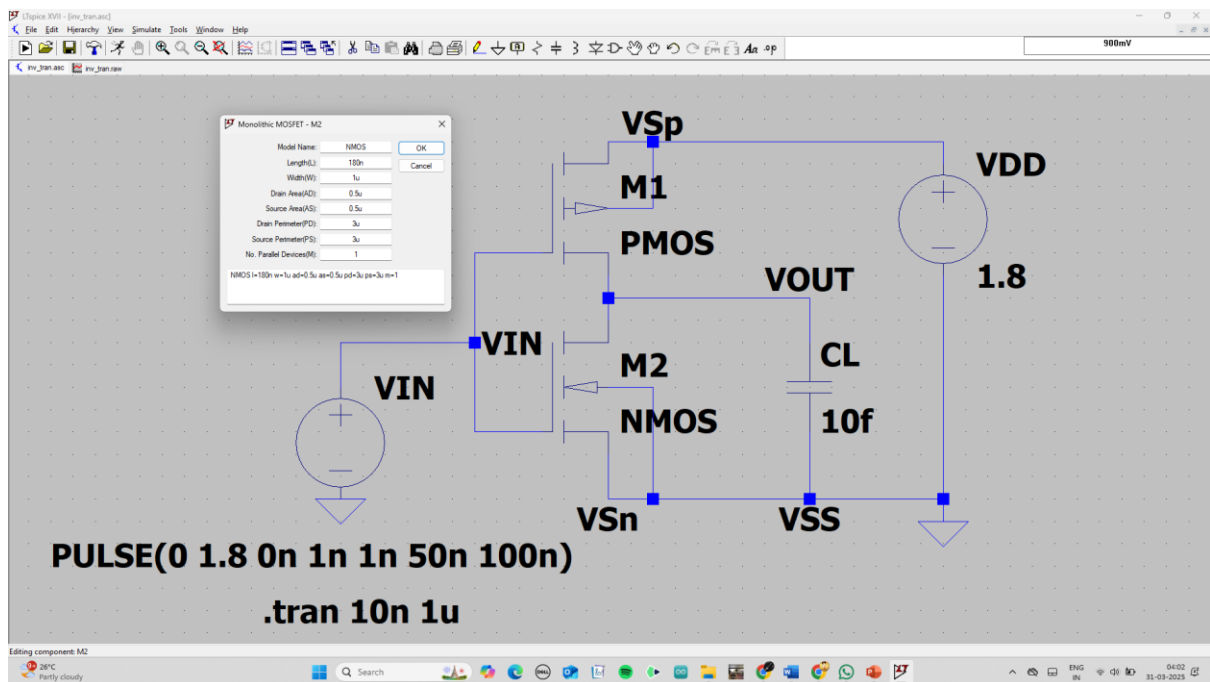
3. Decreased AS, AD, PS, PD (Lower Capacitance & Resistance → Expected to Decrease Delay)			
Parameter	PMOS (M1)	NMOS (M2)	
AS	0.8 μm^2	0.25 μm^2	
AD	0.8 μm^2	0.25 μm^2	
PS	5.0 μm	2.0 μm	
PD	5.0 μm	2.0 μm	

CASE 01 : [INCREASING AS, AD, PS, PD]

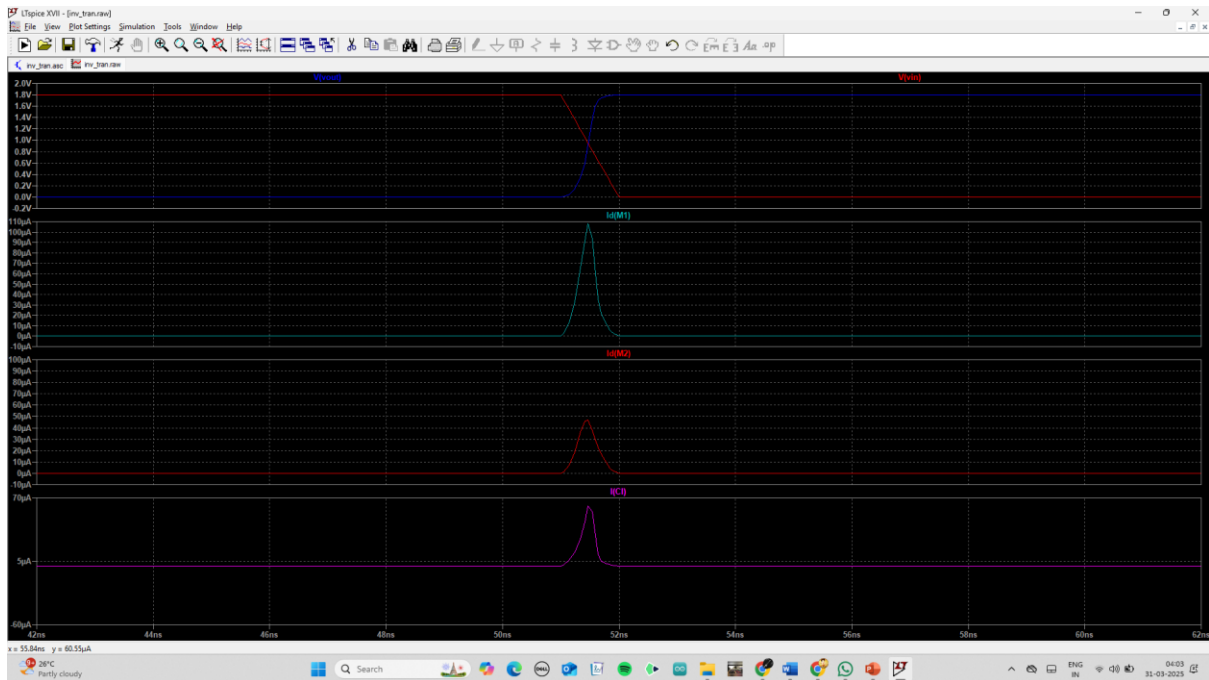
1. PMOS



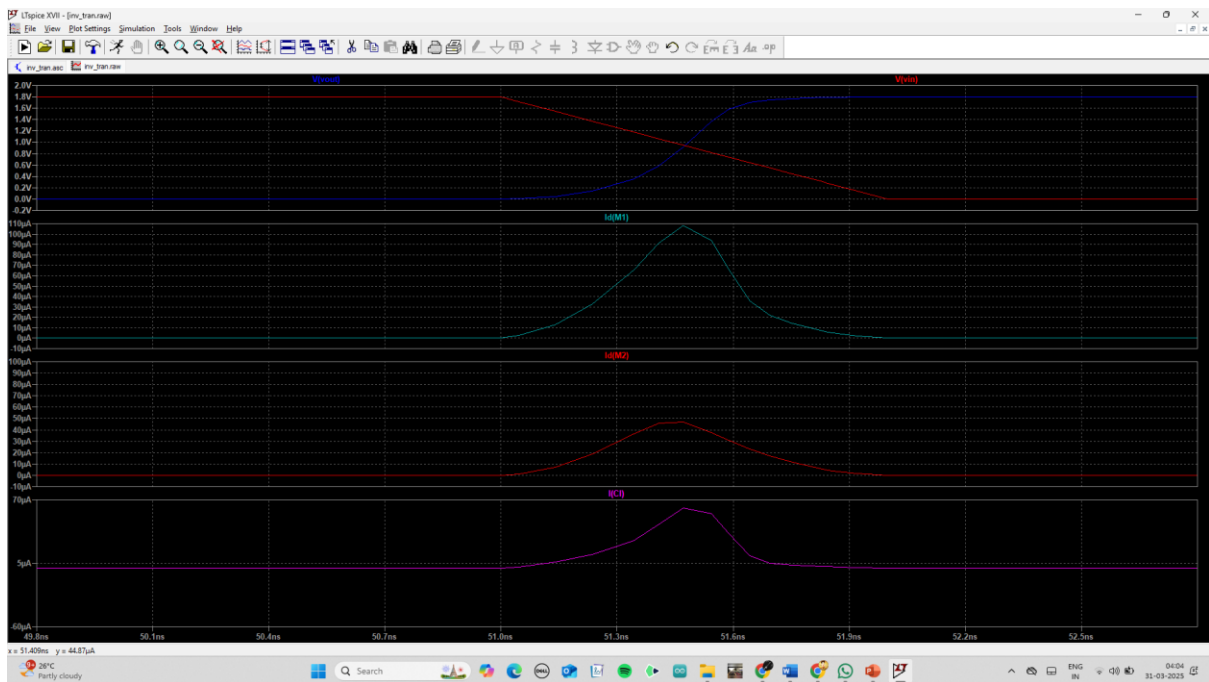
2. NMOS

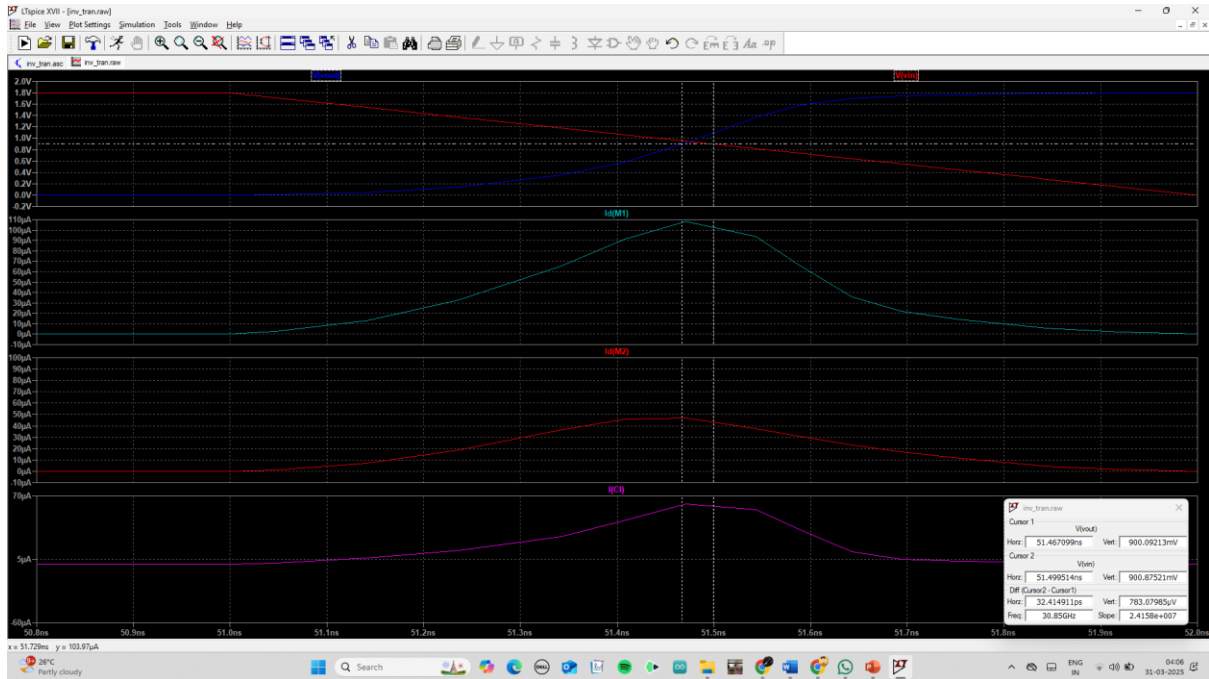


3. $\tau_{\text{ao(PLH)}}$: RISE TIME



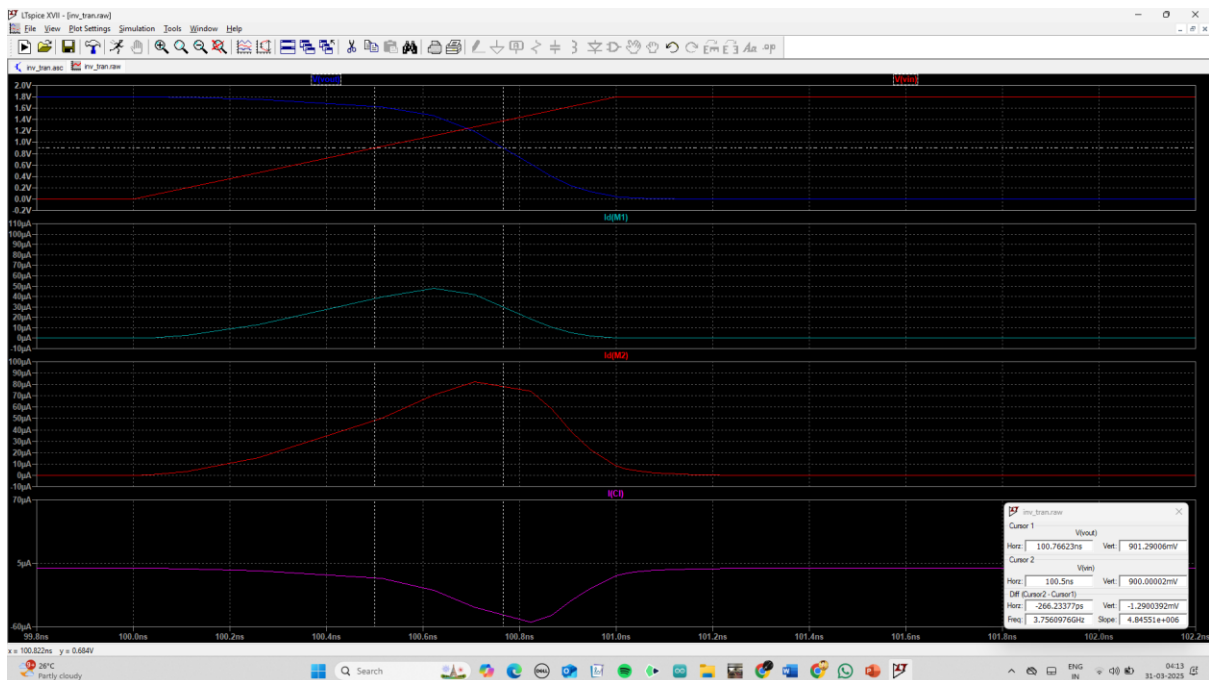
Further zoom ...





$$\text{Tao(PLH)} = 0.032415\text{ns}$$

4. Tao(PHL) : FALL TIME



$$\text{Tao(PLH)} = 0.26623\text{ns}$$

$$5. \text{ P.D} = \text{Tao(PLH)} + \text{Tao(PLH)} / 2 = 0.032415 + 0.26623 / 2 = 0.1493225 \text{ ns} = 149.3225 \text{ ps}$$

Previously , The P.D = Tao (PLH) + Tao (PHL) / 2 = RISE TIME + FALL TIME / 2 = 0.032678ns + 0ns / 2 = 0.016335ns = 16.335 ps .

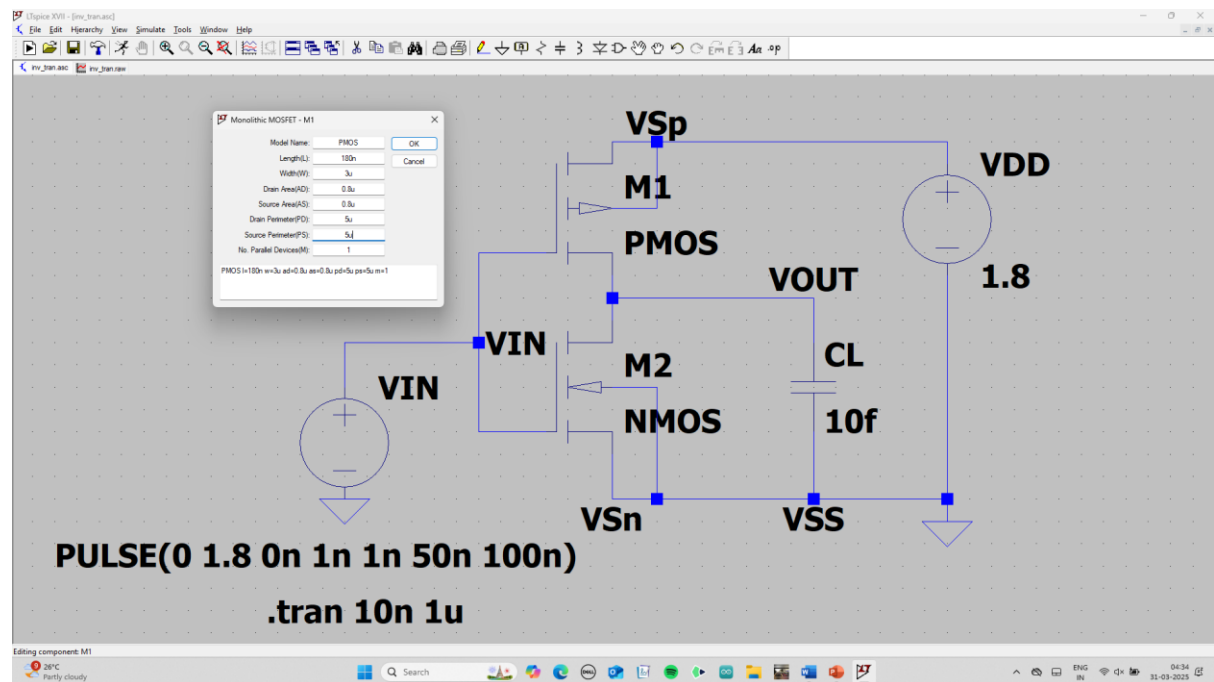
Now after increasing the AS, AD, PS, PD the delay has increased to :
 $(\text{increased S \& D} / \text{previous normal value}) = 149.3225 \text{ ps} / 16.335 \text{ ps} = 9.141 \text{ times . i.e., HUGE !!}$

Henceforth , We conclude that , Delay increases with increase in AS, AD, PS, PD !!

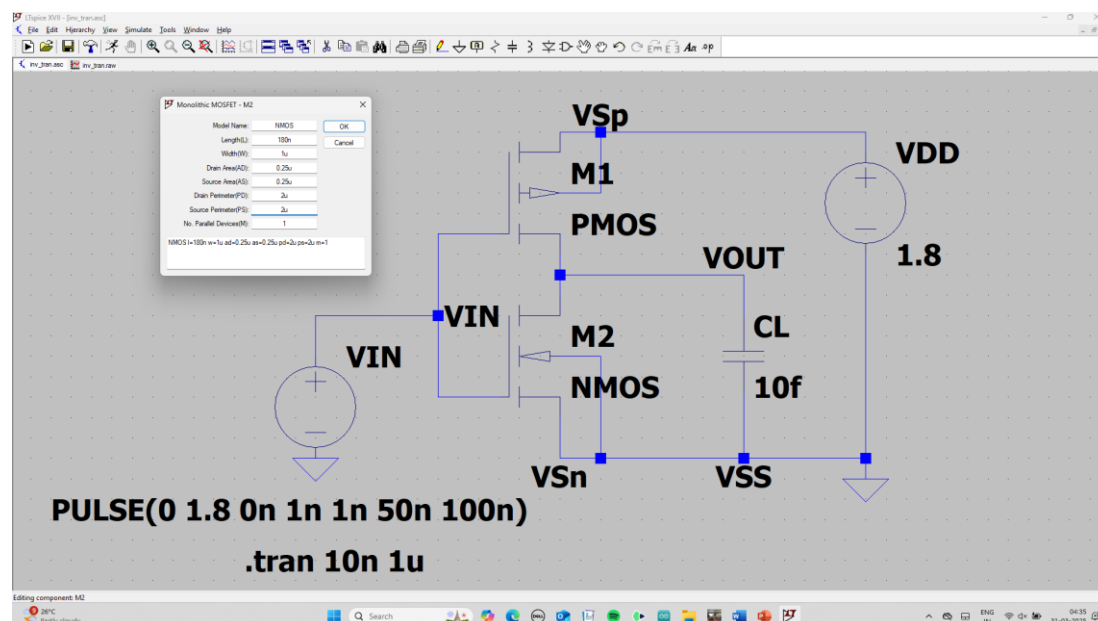
Now , Let's decrease the S & D region Area and Perimeter . . . and see whether the delay decreases or not . . .

CASE 02 : [DECREASING AS, AD, PS, PD]

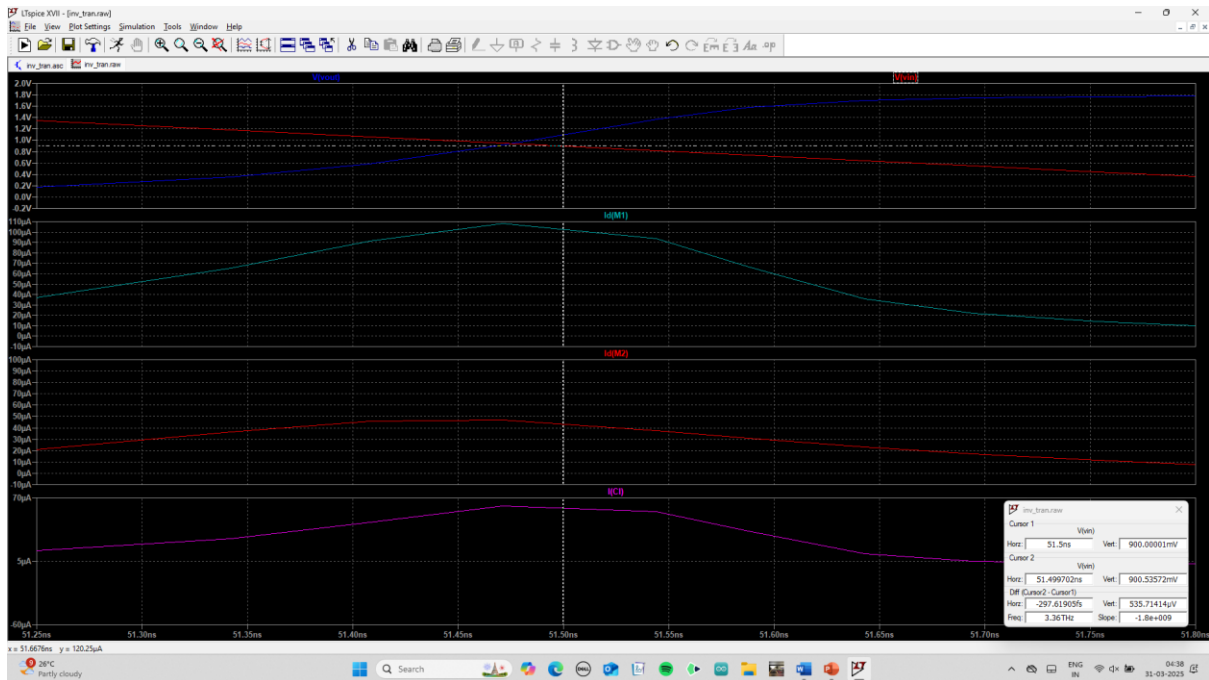
1. PMOS



2. NMOS

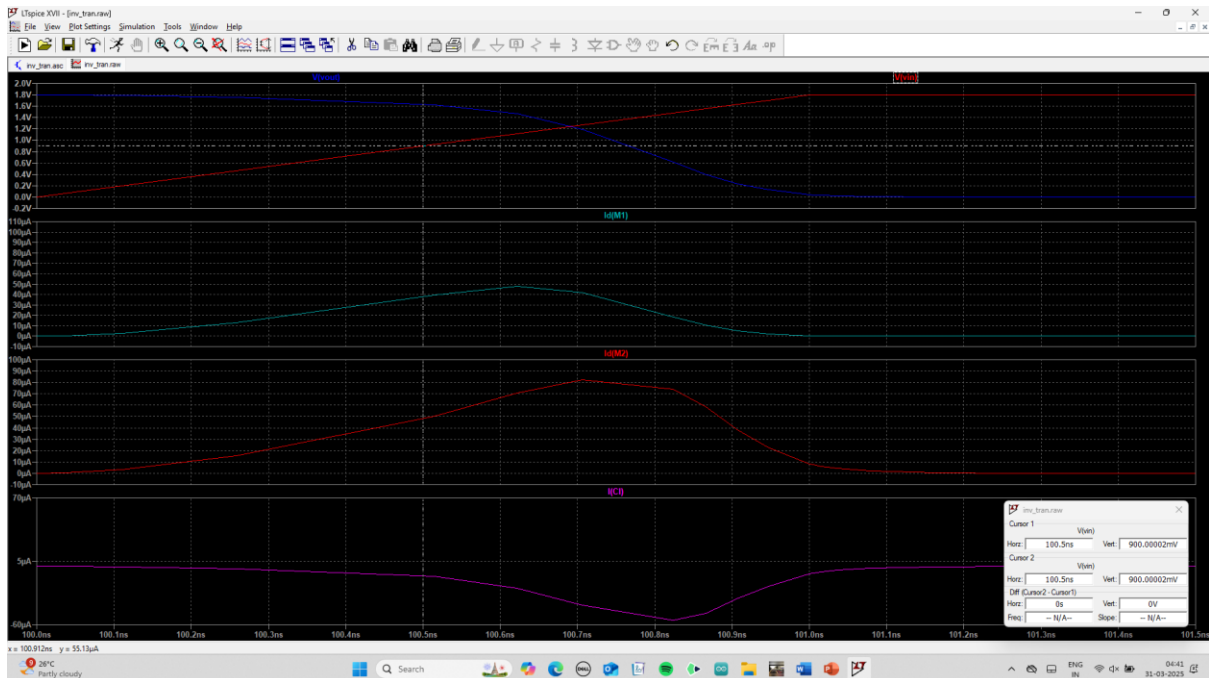


3. $T_{ao}(PLH)$: RISE TIME



297.6190 fs !!!

4. $T_{ao}(PHL)$: FALL TIME



0s

$$5. \quad P.D = T_{ao}(PLH) + T_{ao}(PLH) / 2 = 297.6190 \text{ fs} + 0 / 2 = 148.8095 \text{ fs}$$

So , It's very very clear , Previously , The $P.D = T_{ao}(PLH) + T_{ao}(PHL) / 2 = \text{RISE TIME} + \text{FALL TIME} / 2 = 0.032678\text{ns} + 0\text{ns} / 2 = 0.016335\text{ns} = 16.335 \text{ ps}$.

Now after decreasing the AS, AD, PS, PD the delay has decreased to : **(previous normal value / decreased S & D) = 16.335 ps / 0.1488095 ps = 109.771 times . i.e., VERY VERY VERY VERY SMALL !!**

Henceforth , We conclude that , Delay Decreases with decrease in AS, AD, PS, PD !!